A Scale-Aware Multidomain DETR for Small Object Detection in UAV Remote Sensing Imagery

Jing Li, Yanli Shi, Qihua Hong, and Yi Jia

***Abstract*—Object detection in unmanned aerial vehicle (UAV) remote sensing imagery is significantly challenged by small-scale objects, dense distributions, and complex backgrounds. Existing detection transformer (DETR)-based models struggle with multiscale feature extraction, and traditional multihead self-attention (MHSA) often introduces redundancy and noise when handling high-frequency (HH) details. Moreover, current methods lack effective multidomain feature modeling. To address these issues, this article proposes a scale-aware multidomain DETR (SME-DETR) for small object detection in UAV remote sensing imagery. First, a scale-aware attentional backbone (SAA-Backbone) is designed, where the dynamic scale-aware (DSA) module adaptively fuses multiscale features using depthwise separable convolutions and dynamic weighting. In addition, a directional channel attention (DCA) module further enhances edge and texture representation. Then, a multidomain augmented pyramid (MDAP) is constructed, integrating a CSP-based multidomain kernel (CSP-MDKernel) to jointly optimize spatial, frequency, and channel domain features. A spatial-channel fusion convolution (SCFConv) is employed to preserve fine-grained details. Finally, an efficient attention-based intrascale feature interaction (EfficientAIFI) module is proposed, which replaces the complex matrix operations of traditional MHSA with linear element-wise multiplication. At the same time, it maintains global dependencies through normalized inner products between queries and keys. In addition, a lightweight background-suppression gate (BSG) is incorporated to mitigate false positives induced by cluttered backgrounds, further enhancing detection robustness. Experiments on the VisDrone-DET, RSOD, and SeaDronesSeeV2 datasets with the core SME-DETR framework demonstrate that SME-DETR achieves mAP@0.5 scores of 53.0% and 97.8% on the VisDrone-DET and RSOD, respectively, improving the RT-DETR-r18 baseline by 6.0% and 3.0%, respectively. Moreover, SME-DETR significantly outperforms most state-of-the-art detectors.**

***Index Terms*—Attention mechanism, multidomain enhancement, multiscale feature fusion, small object detection.**

I. Introduction

IDENTIFYING and locating small objects in remote sensing images presents significant technical challenges. Under aerial perspectives, objects are typically small, densely distributed, and lack sufficient detail representation, making them more susceptible to interference from complex backgrounds. Specifically, small object detection faces three major challenges: 1) significant scale variations often lead to localization errors; 2) low resolution and complex background interference reduce feature extraction efficiency, thereby severely degrading detection performance; and 3) small objects are susceptible to feature degradation, where even slight feature loss can cause a drastic drop in accuracy. In addition, UAV remote sensing imagery is often characterized by multiscale variations, dynamic environments, and long-tailed data distributions, which further constrain the feature representation capabilities of traditional detection methods, particularly in small object detection tasks. With the continuous improvement in the quality and quantity of remote sensing imagery, object detection has found expanding applications in military reconnaissance [1], traffic monitoring [2], and disaster response [3], leading to increasingly stringent requirements for detection accuracy.

Currently, unmanned aerial vehicle (UAV) aerial image object detection predominantly relies on deep learning architectures and is generally divided into two mainstream approaches. The first category is two-stage detection methods, such as Cascade region-based convolutional neural network (R-CNN) [4], Faster R-CNN [5], and Mask R-CNN [6]. These methods generate candidate region proposals using a region proposal network (RPN), then perform object localization through classification and bounding box regression. Although these methods generally achieve high detection accuracy, their complex computational processes make it challenging to meet real-time inference requirements. The other is one-stage detection methods, including the YOLO series [7], SSD [8], EfficientDet [9], and RetinaNet [10]. These methods employ end-to-end convolutional neural networks to directly predict object locations and categories, eliminating the need for region proposal generation. This design improves inference speed while maintaining competitive detection accuracy, meeting real-time processing requirements. Beyond these mainstream approaches, Zhou et al. [11] proposed MaDiNet, a hybrid framework that integrates diffusion models with Mamba—a state space modeling technique—for SAR target detection. Unlike conventional anchor-based or anchor-free detectors, MaDiNet formulates detection as a generative process that predicts bounding box centers and sizes, thereby improving performance in complex backgrounds. Although designed for SAR imagery,

Received 27 May 2025; revised 5 September 2025 and 16 October 2025; accepted 18 October 2025. Date of publication 23 October 2025; date of current version 17 November 2025. This work was supported in part by Jilin Province Science and Technology Department Project under Grant YDZJ202201ZYTS602, in part by Inner Mongolia Autonomous Region Science and Technology Department Project under Grant 2023YFHH0024, and in part by Jilin Provincial Department of Education Scientific Research Project under Grant JJKH20251305KJ. *(Corresponding author: Yanli Shi.)*

Jing Li, Qihua Hong, and Yi Jia are with the College of Information and Control Engineering, Jilin Institute of Chemical Technology, Jilin City 132022, China (e-mail: lijing64@jlict.edu.cn; hongqihua@jlict.edu.cn; jy1161786040@163.com).

Yanli Shi is with the College of Science, Jilin Institute of Chemical Technology, Jilin City 132022, China (e-mail: syl@jlict.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3624765

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

its diffusion-state space model (SSM) paradigm provides valuable insights for object detection in aerial scenarios with high background interference. In recent years, transformer-based object detection frameworks have demonstrated distinct advantages. Leveraging the self-attention mechanism to model long-range dependencies effectively captures complex object semantic relationships. For example, vision transformer (ViT) [12] pioneered the application of transformers in visual classification tasks, achieving performance superior to traditional CNNs when trained on large-scale datasets. Detection transformer (DETR) [13] further integrates CNN's local feature extraction strengths with transformers' global modeling capabilities. It adopts an end-to-end learning strategy that removes the reliance on traditional anchor box design and omits postprocessing steps like nonmaximum suppression (NMS). However, the substantial computational burden introduced by the transformer structure in DETR limits its real-time detection capabilities. To address this issue, Zhao et al. [14] introduced real-time detection transformer (RT-DETR), outperforming existing YOLO series models in precision and inference speed. Building upon this foundation, several domain-specific variants have been proposed. For instance, Ma et al. [15] developed OWRT-DETR for open-water search and rescue, which incorporates lightweight modules to strengthen weak small-object features and enhance cross-scale interactions, improving detection accuracy while maintaining real-time efficiency. More recently, Zhong et al. [16] introduced predator conv, scalewise differential module, whale particle optimization (PSWEP)-DETR, a remote sensing-oriented framework that integrates predator convolution (PdConv), a scale-wise differential module (SDM), and whale particle optimization (WPO) to improve geometric adaptability and multiscale feature representation. However, its high complexity and large parameter count require substantial hardware resources.

Given its strong balance between accuracy and speed, this article adopts RT-DETR-r18 as the baseline and further enhances it to address the limitations in detecting small objects under complex UAV scenarios. The RT-DETR model adopts a convolutional neural network (CNN)-based backbone (e.g., ResNet [17]) for feature extraction and introduces an attention-based intra-scale feature interaction (AIFI) module at the final stage. Leveraging MHSA enhances intrascale feature interactions, thereby improving the capture of global contextual information. In the neck network, the model integrates a cross-scale feature fusion module (CCFM) module inspired by the path aggregation feature pyramid network (PAFPN) [18], enabling feature map integration via bidirectional information flow. After a query selection mechanism (selecting the top-K items), the fusion module generates dense predictions, which serve as initial object query input to the decoder for final detection. However, in UAV aerial image tasks, ResNet's fixed convolutional kernels and uniform feature extraction strategy limit its capability to detect multiscale objects, leading to suboptimal performance, especially in the presence of small objects and complex backgrounds. In the AIFI module, the matrix multiplication operations associated with self-attention introduce considerable computational complexity and training overhead. Meanwhile, in the neck network, RT-DETR employs simple residual connections and RepBlock [19] for multiscale feature fusion. However, standard convolutional operations exhibit inherent limitations when dealing with variations in object distribution and scale caused by different shooting angles. This makes it difficult to capture and represent complex feature information sufficiently.

The challenges above underscore the necessity for three fundamental capabilities in UAV-based small object detection: 1) adaptive multiscale and directional feature modeling, to preserve edge sharpness and texture integrity across a wide range of object sizes; 2) enriched multidomain representation, to effectively suppress background interference and mitigate interclass ambiguity; and 3) efficient global context modeling, to reduce computational redundancy while maintaining robust semantic expressiveness. Guided by these requirements, a scale-aware multidomain DETR (SME-DETR) for small object detection in UAV remote sensing imagery is proposed. Built upon the RT-DETR-r18 architecture, the model integrates the advantages of CNNs and transformers. First, a scale-aware attentional backbone (SAA-Backbone) is designed by integrating a novel dynamic scale-aware (DSA) module and a directional channel attention (DCA) module, effectively boosting the model's representation of multiscale objects while improving its sensitivity to fine-grained features. Then, a multidomain augmented pyramid (MDAP) is constructed by introducing the newly designed CSP-multidomain kernel (MDKernel) and spatial-channel fusion convolution (SCFConv), enabling dynamic collaborative optimization across spatial, frequency, and channel domains. This design significantly enhances the identification of small objects in cluttered environments. Finally, an efficient attention-based intrascale feature interaction (EfficientAIFI) module is proposed, which simplifies the interaction process to reduce redundant computations and optimizes element-wise information fusion to suppress noise interference, thereby improving detection accuracy. To further bolster the framework's robustness against complex backgrounds, a lightweight background-suppression gate (BSG) is also incorporated as an auxiliary component. The main contributions of this work can be summarized as follows.

1) An SAA-Backbone is designed to enhance multiscale and directional feature modeling. It integrates multiscale depthwise convolutions, dynamic weighting, and DCA to effectively capture edge and texture information across varying object sizes and orientations.
2) An MDAP is constructed by combining a CSP-based MDKernel (CSP-MDKernel) and an SCFConv. These two modules work collaboratively to enrich multiscale and multidomain representations while preserving fine-grained features under complex backgrounds, thereby improving small object detection performance.
3) An EfficientAIFI module is proposed to structurally streamline attention computation by removing the key–value interaction and preserving only the query–key relation. This design maintains contextual modeling capacity and improves detection accuracy.
4) The proposed SME-DETR is experimentally validated on the VisDrone-DET, RSOD, and SeaDronesSeeV2 datasets. Experimental results show that the model exceeds the baseline and several state-of-the-art object

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

detection methods in terms of detection performance. Furthermore, visualization analyses demonstrate SME-DETR's advanced ability to detect objects in complex scenarios.

In addition, a BSG is incorporated into the framework as an auxiliary enhancement. It complements the three main modules by specifically addressing background-induced false activations. Although designed as an auxiliary component rather than a core contribution, BSG consistently reduces false positives with negligible computational overhead, as shown in Section IV.

This article is organized as follows. Related work is reviewed in Section II, followed by the proposed methodology in Section III. Experimental results and comprehensive analysis appear in Section IV, while conclusions and research outlook are provided in Section V.

II. Related Works

A. Feature Extraction Methods

The diversity of object scales, arbitrary orientations, and complexity of backgrounds in drone remote sensing data pose significant challenges. This demands feature extraction techniques that possess strong multiscale perception capabilities and enable precise object detection in complex backgrounds. Traditional CNNs typically use fixed-size convolutional kernels to extract local features. However, this fixed kernel and uniform feature extraction approach often struggles to adapt to the diversity of objects at different scales in complex backgrounds. Many studies have proposed optimization methods for multiscale features and complex backgrounds. For example, Nie et al. [20] developed a novel framework incorporating the enhanced context module (ECM) and ternary attention module (TAM). The ECM enriches the contextual information of shallow features, while the TAM selectively fuses useful contextual information through adaptive refinement, jointly improving the performance of multiscale object detection. Singh et al. [21] proposed a heterogeneous convolution method called HetConv, which integrates the advantages of standard convolution and depthwise separable convolution. By introducing convolution kernels of diverse sizes and shapes, it captures multiscale information while using grouped convolution techniques to reduce computational complexity effectively. Hou et al. [22] proposed an attention-enhanced end-to-end discriminative network named TDO-Net, which incorporates multiscale dilated convolution modules into ResNet-50 to enlarge the receptive field and capture multiscale features. Integrating a triplet attention module strengthens the network's emphasis on critical features. Han et al. [23] proposed a novel method named multi-scale spatial supervision convolutional network (MSSCN), which integrates Gaussian kernel functions with a multiscale spatial attention mechanism to accurately detect standing dead trees affected by pine wilt disease in complex scenes. Ran et al. [24] proposed a model named MSCCA, which employs PeleeNet as the backbone network and enhances channel attention in feature maps while integrating multiscale contextual information to improve recognition precision for oriented targets in aerial imagery.

Although current methods have significantly progressed in extracting multiscale features and addressing complex scenes, most still lag in modeling directional features, particularly in handling small objects and directional characteristics under complex backgrounds. To address this issue, this article proposes a novel SAA-Backbone, which incorporates a dual-path fusion (DPF) module and introduces two innovative components: a DSA module and a DCA module, aiming to enhance the representation of features across multiple scales and directional modeling. Specifically, the DSA module adaptively adjusts the contributions of convolutional kernels across diverse receptive fields, thereby enhancing the network's capability to extract multiscale object features precisely. Meanwhile, the DCA module enhances the perception of local image details and improves the modeling of directional features in complex scenes.

B. Small Object Detection Techniques

The core challenges of small object detection lie in the minute object sizes (typically smaller than 32×32 pixels), loss of texture due to low-resolution imaging, and background clutter interference. To address these issues, researchers have proposed various optimization approaches, broadly categorized into two main directions: feature fusion optimization and feature enhancement. Regarding feature fusion optimization, the earliest proposed feature pyramid network (FPN) [25] provided an efficient design for feature interaction, laying the foundation for subsequent studies. Since then, several improved modules have been successively proposed, including the path aggregation network (PANet) [18], bidirectional FPN (Bi-FPN) [9], adaptively spatial feature fusion (ASFF) [26], and neural architecture search FPN (NAS-FPN) [27]. Meanwhile, Zhang et al. [28] proposed a convolutional neural network model named MFR-CNN, enabling more accurate identification of occluded and small targets by leveraging multiscale features and global contextual information. In addition, Woo et al. [29] introduced an object detection framework named StairNet, which centers on a feature combination module that progressively aggregates semantic information from multiscale feature maps within the CNN. This design enhances the semantic representation of low-level features and reduces the semantic gap between different layers. Deng et al. [30] developed the extended FPN (EFPN), which builds upon the traditional FPN by adding a high-resolution pyramid level. In addition, a feature texture transfer (FTT) module is designed to generate features containing richer details in small object regions, thereby improving small object detection effectiveness. In terms of feature enhancement, Cheng et al. [31] proposed a novel feature enhancement network (FENet), which integrates a context feature enhancement (CFE) module and a dual attention feature enhancement (DAFE) module into the Faster R-CNN framework to improve object detection performance in remote sensing images. Zhang and Shen [32] introduced the multistage feature enhancement pyramid network (Multistage FEPN), which generates multistage feature maps, fuses adjacent feature maps, and further enhances detection precision in optical remote sensing images through a feature enhancement module (FEM). Meanwhile, Zhang et

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

al. [33] proposed an improved backbone network, MHSA-Darknet, incorporating a multihead self-attention (MHSA) module to retain rich global context and extract more distinctive representations, enhancing object detection capabilities.

However, existing methods still exhibit certain limitations. Most feature fusion approaches fail to effectively coordinate global and local information, thereby compromising the preservation of detailed information in small object regions. Feature enhancement methods predominantly rely on attention mechanisms, which often struggle to distinguish objects from background noise in complex scenes. Moreover, current methods mainly concentrate on spatial domain optimization while neglecting the potential of frequency domain information in enhancing the discriminability of small objects. To address these issues, this article proposes an MDAP structure. MDAP integrates an SCFConv module with a multidomain feature fusion strategy to strengthen object features and edge details, suppress background interference, and enhance global contextual awareness. The proposed approach effectively improves small object recognition performance and stability by facilitating information interaction across spatial, frequency, and channel domains.

C. Attention Mechanism

Neural networks in visual recognition tasks utilize attention mechanisms to focus on critical regions of the input data by adjusting the importance of different spatial locations or feature channels. This enhances the feature extraction capability, leading to better performance in terms of the accuracy and robustness of the model. In practice, multiple forms of attention mechanisms have been developed. For example, the spatial transformer network (STN) [34] applies spatial attention to model the relationships between different regions in an image, thereby determining the importance of each location. This helps the model focus on object areas while reducing background interference. Subsequently, the squeeze-and-excitation network (SENet) [35] introduced channel attention by assigning different weights to the channels of the input feature maps, enhancing focus on informative channels while suppressing less relevant ones, thus improving the capture of object features and contextual information. The convolutional block attention module (CBAM) [36] combines spatial and channel attention, enabling the model to simultaneously attend to key information across spatial and channel dimensions for more comprehensive feature extraction. Song et al. [37] proposed a multiscale attention-based deep neural network (MSA-DNN), which leverages global semantic information from image-level annotations to guide the detection task and constructs an innovative multiscale attention module (MSAM).

In recent years, self-attention-based feature interaction mechanisms have demonstrated significant advantages in visual tasks. The classic MHSA mechanism [38] captures long-range contextual information by computing interaction weights among queries, keys, and values. However, its computational complexity grows quadratically with the sequence length, making it inefficient for high-resolution image detection tasks. To address this issue, deformable DETR [39] adopts a deformable attention mechanism that dynamically samples key positions, reducing the computational complexity to $O(NKd)$, where N denotes the number of queries, K is the number of sampled points per query, and d is the feature dimension. However, its sparse sampling strategy may miss the edge features of small objects. Linformer [40] reduces the complexity to $O(Nd^2)$ by projecting the key–value matrix into a low-dimensional space. Still, this low-rank compression can lead to the loss of high-frequency (HH) details such as edges and corners. Performer [41] employs random orthogonal feature mappings to replace dot-product attention, achieving linear complexity $O(N)$. Still, the randomness in projection undermines the interpretability of attention weights and poses challenges for complete optimization in hardware deployment. It is worth noting that most existing methods retain the traditional query–key–value (QKV) interaction structure. For small object detection, the key vectors typically dominate spatial localization, while the value vectors play a relatively minor role in semantic enhancement. This reveals the redundancy of the traditional QKV structure—the fixed full interaction pattern results in a large amount of redundant computation. Particularly when handling HH details, the redundant key–value projection layers (such as the linear transformation matrices in MHSA) may introduce noise interference, leading to reduced detection accuracy.

To address this issue, this article proposes an EfficientAIFI module. This module adopts efficient additive attention (EAA) [42] to replace the traditional MHSA, relying solely on query–key interactions to model token relationships. This design avoids unnecessary key–value computation overhead while effectively preserving critical information.

III. Methods

A. Overall Structure

The proposed SME-DETR model, designed for small object detection in UAV remote sensing imagery, is illustrated in Fig. 1. First, an SAA-Backbone is constructed to replace the original ResNet18. By integrating a DPF module that combines a DSA module and a DCA module, the network achieves an adaptive fusion of multiscale features and enhancement of directional textures, significantly improving the ability to capture fine details of small objects in remote sensing imagery. Second, an MDAP is designed to optimize multiscale feature fusion through the multibranch structure and a frequency–spatial attention mechanism of the CSP-MDKernel module. In addition, an SCFConv is applied to the high-resolution P2-level features to enhance fine-grained details and perform contextual modeling, effectively mitigating the interclass similarity between small objects and complex backgrounds in remote sensing imagery. Finally, to address the potential noise interference caused by redundant key–value projections, an EfficientAIFI module is proposed. This module retains only the interaction between queries and keys, enabling efficient global relation modeling through linear element-wise operations. In addition, a BSG is incorporated within the decoder to mitigate false positives by adaptively penalizing detections in background-dominated regions, enhancing overall robustness in cluttered scenarios.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

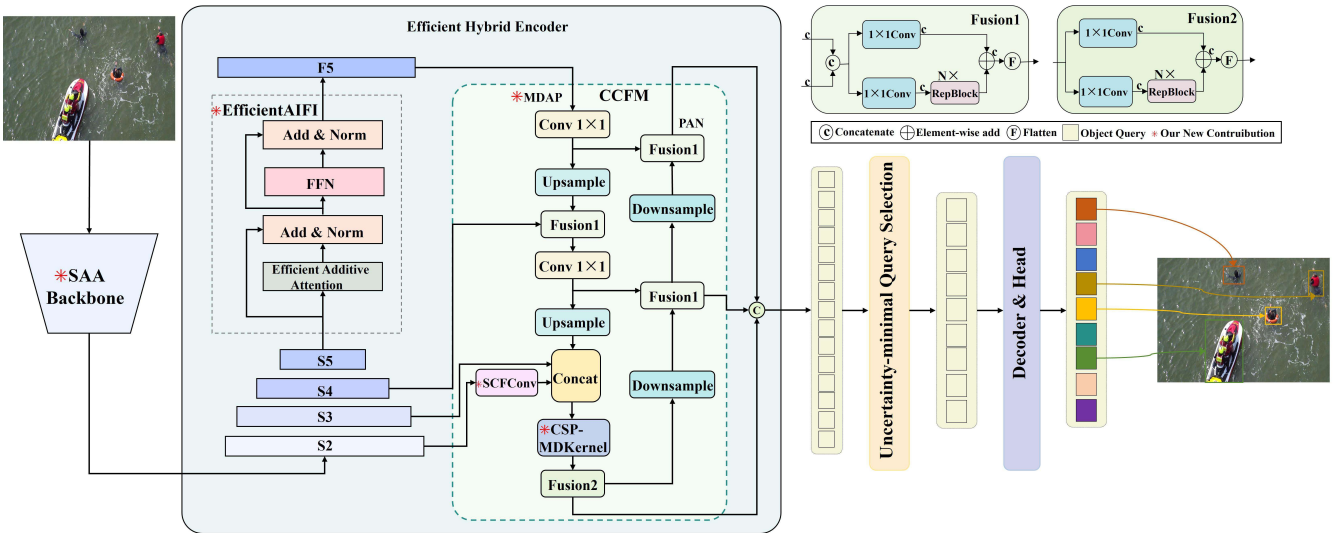


Fig. 1. Overall architecture of the proposed SME-DETR. The model consists of three major components: 1) SAA-Backbone, which extracts multiscale and directional features through the integration of DSA and DCA modules; 2) efficient hybrid encoder, which incorporates EfficientAIFI to enhance spatial feature representation, while MDAP performs multidomain feature fusion in the spatial, frequency, and channel domains via CSP-MDKernel; and 3) decoder and head. Initial object queries are selected via an IoU-aware mechanism to initialize the decoder, which subsequently performs iterative refinement of the predicted bounding boxes and confidence scores in the auxiliary prediction head. The BSG acts on the classification scores to suppress false activations.

B. Scale-Aware Attentional Backbone

Traditional CNNs face several critical challenges during the feature extraction stage as follows.

1) They rely on fixed-size convolutional kernels, which limit adaptability to multiscale objects, resulting in the loss of fine details in small objects and insufficient capture of global contextual information.
2) The initial convolutional layers typically perform simple feature fusion, lacking dynamic adjustment and adaptability. They cannot dynamically optimize the convolutional kernel's weight based on the varying features of the input image, leading to reduced detection accuracy in complex scenes.
3) Traditional CNNs primarily rely on global convolution operations for directional feature modeling. However, this approach often neglects the details of local directional variations, resulting in inadequate extraction of edge and texture features.
4) As network depth increases, the computational complexity and number of parameters rise sharply. This poses significant challenges regarding computational efficiency and real-time performance, particularly on resource-constrained devices.

To address these issues, this study proposes an innovative backbone network—SAA-Backbone, as shown in Fig. 2(a) and (b). This network incorporates an innovative DPF Module, shown in Fig. 2(c). The DPF consists of two submodules: the DCA module [see Fig. 2(d)] and the DSA module [see Fig. 2(e)]. By adaptively fusing the outputs of these two submodules, the SAA-Backbone enhances the modeling capability of multiscale information and improves the extraction of directional features, thereby boosting small object detection performance.

DSA is designed to adaptively extract features at different scales and fuse them through a dynamic weighting mechanism. This module has two main components: 1) **multiscale depthwise separable convolution:** the input feature map x is processed using multiple depthwise convolution kernels of varying scales to extract spatial features, resulting in a set of multiscale feature maps $x_1, x_2, \ldots, x_n$. 2) **Dynamic weighting mechanism:** to evaluate the relative importance of features at different scales, DSA introduces a set of learnable weight parameters $w = [w_1, w_2, \ldots, w_n]$, which are normalized using the softmax function. These weights are continuously optimized during training via backpropagation to meet the requirements of different tasks, enabling the model to adjust each convolution kernel's contribution dynamically. Finally, the outputs of the multiscale convolutions are fused via a weighted summation to form the integrated feature map x_{out}, which is then passed through a pointwise convolution (1×1 convolution) to further mix and transform channel information, thereby enhancing feature representation capability.

The calculation of x_{out} is given as follows:

$$x_{\text{out}} = \sum_{i=1}^{n} w_i \cdot x_i. \tag{1}$$

Notably, the multiscale convolutions in the DSA module adopt depthwise separable convolutions, significantly reducing the number of parameters and computational burden compared to standard convolutions. Moreover, the dynamic weighting mechanism is implemented via a limited set of shared, learnable scalar weights normalized by a softmax function, thus avoiding the overhead of fully connected layers or matrix multiplications.

This design is well-suited for UAV imagery, where small targets often exhibit diverse texture characteristics. Fine details, such as runway lines or rooftop edges, require small receptive fields (e.g., 3×3), whereas low-frequency (LL) structures

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

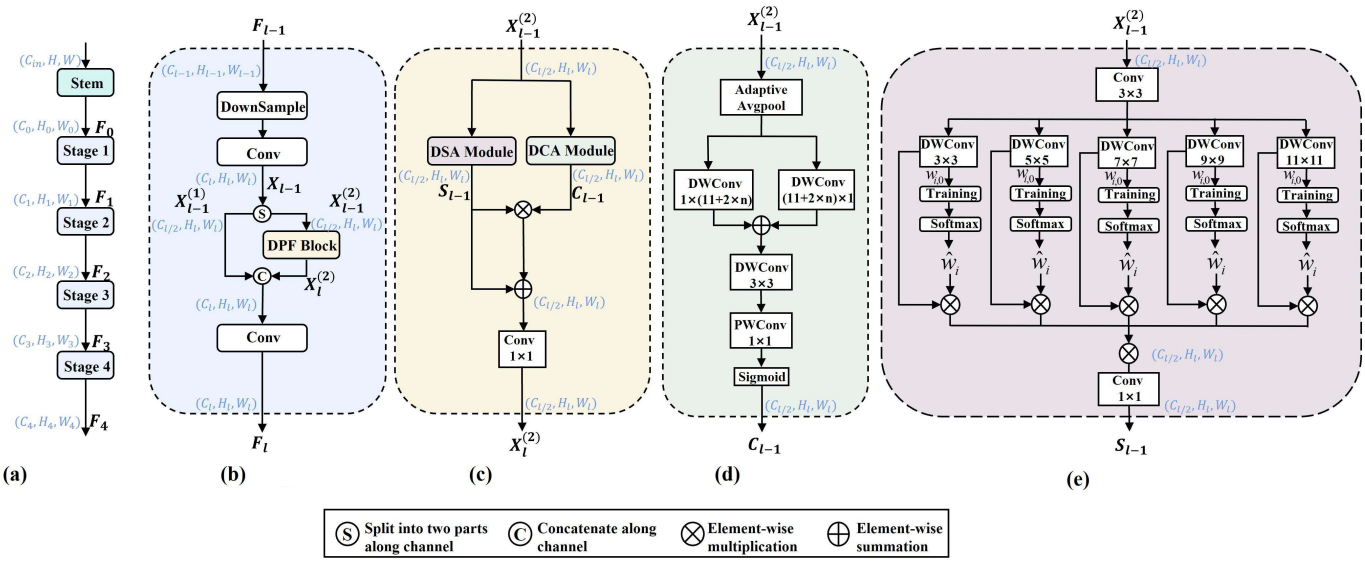


Fig. 2. Detailed architecture of the SAA-Backbone. (a) and (b) Four-stage backbone and feature downsampling process. (c) DPF block, which integrates (d) DCA module for enhanced directional texture and edge modeling through vertical and horizontal convolutions and (e) DSA module for adaptive multiscale feature fusion.

like tank outlines benefit from larger kernels (e.g., 7×7). The DSA module addresses this through multiscale depthwise convolutions combined with dynamically learned weights. For smaller, edge-rich targets (e.g., pedestrians), more weight is allocated to 3×3 filters; for larger structures, intermediate-scale filters (e.g., 5×5) better balance local detail and global shape. This adaptability allows the network to capture intricate textures while maintaining structural consistency across varying object sizes and scene complexities.

DCA enhances the capacity to capture directional information such as edges and textures. Its implementation process is as follows.

1) **Global Information Extraction:** The input feature map $X \in \mathbb{R}^{C\times H\times W}$ (where C is the number of channels, H is the height, and W is the width) is passed through an adaptive global average pooling (AGAP) layer to extract global contextual information by computing the global average value of each channel [as shown in (2)].

2) **Directional Feature Enhancement:** The pooled feature map is separately convolved using horizontal convolution Conv_h with a size of $1 \times (11+2 \times n)$ and vertical convolution Conv_v with a size of $(11 + 2 \times n) \times 1$, generating two directional attention maps [as shown in (3)].

3) **Fusion and Nonlinear Transformation:** The horizontal and vertical attention maps are combined and processed using a depthwise separable convolution to refine the features while reducing computational cost and parameter overhead, thereby enhancing representational capability.

4) **Channel Attention Generation:** Finally, the fused directional features are normalized using a Sigmoid activation function to produce the final channel attention map A [as shown in (4)]

$$\text{AGAP}(X)_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_c(i, j), \quad c = 1, 2, \ldots, C \quad (2)$$

where $X_c(i, j)$ represents the value of the c channel at position (i, j) in the input feature map X

$$h_{\text{attn}} = \text{Conv}_h(\text{AGAP}(X)), \quad v_{\text{attn}} = \text{Conv}_v(\text{AGAP}(X)) \quad (3)$$

$$A = \text{Sigmoid}(\text{PointwiseConv}(\text{DWConv}(h_{\text{attn}} + v_{\text{attn}}))). \quad (4)$$

In contrast to conventional isotropic convolutions, the DCA module employs vertical and horizontal filters to selectively enhance direction-specific edge responses. This design is particularly effective for UAV imagery, where small objects often exhibit weak and sparse contours. By aggregating directional information, DCA strengthens edge continuity and reduces ambiguity in object boundaries. In addition, the integrated AGAP mechanism captures global structural statistics, guiding the attention focus toward dominant structural directions while suppressing background noise, such as vegetation or water textures.

The SAA-Backbone further benefits from the synergy between DCA and DSA. The directional attention maps from DCA guide the multiscale fusion in DSA, allowing the network to emphasize directionally aligned features across scales. For instance, in the SeaDronesSeeV2 dataset, DCA enhances vertical contours of swimmers, while DSA leverages 3×3 filters to preserve fine-grained limb details and 7×7 filters to capture holistic body structures. This coordinated mechanism ensures a comprehensive representation that retains edge clarity and texture completeness. Furthermore, DCA's global attention modulation suppresses background clutter, while DSA's dynamic scale selection mitigates overfitting to noisy or irrelevant textures. However, the DCA module is limited to enhancing features in horizontal and vertical directions. Its performance may degrade under severe rotation or tilt, highlighting the need for orientation-invariant modeling mechanisms in future work.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

In summary, the proposed SAA-Backbone effectively mitigates the computational and parameter inefficiencies commonly encountered in traditional CNNs. By integrating depthwise separable convolutions and a lightweight dynamic weighting strategy within the DSA module, our design significantly reduces model complexity while preserving feature expressiveness. In addition, the DCA module leverages directional convolutions guided by global contextual cues and refines features using depthwise separable convolutions, ensuring efficient modeling of edge and texture structures. The effectiveness of these architectural improvements is validated in Section IV-D, where experimental results demonstrate notable reductions in parameter count and GFLOPs compared to traditional CNN-based backbones.

C. Multidomain Augmented Pyramid

The P2 layer is located in the shallower part of the network. Its higher resolution allows for the extraction of detailed features, a key factor in recognizing the shape and texture of small objects. Therefore, traditional small object detection methods often incorporate the P2 layer in the FPN to enhance detection performance. However, this strategy has several limitations as follows.

1) The P2 layer has a relatively small receptive field and strong local dependency, making it challenging to model long-range relationships. This leads to insufficient contextual representation for small objects and increases susceptibility to background interference.
2) Although the P2 layer retains rich spatial details (e.g., edges and textures), it lacks high-level semantics and global context, exacerbating interclass similarity between small objects and background and compromising detection accuracy.
3) The high-resolution P2 layer substantially increases the number of candidate boxes. In anchor-based detectors, this leads to the computational complexity of NMS growing quadratically. Although DETR-based models eliminate the need for NMS, they still suffer from a considerable computational burden, adversely affecting inference efficiency.

To address the above issues, this article proposes an MDAP whose core innovation lies in the design of two novel modules: the CSP-MDKernel [as shown in Fig. 3(a)] and the SCFConv [as shown in Fig. 3(d)]. Specifically, the CSP-MDKernel adopts a feature partitioning strategy to enhance multiscale features' processing and representation capabilities. At the same time, the SCFConv improves feature diversity and interaction through channel reorganization and concatenation mechanisms.

1) CSP-MDKernel: Enhancing Multiscale and Multidomain Feature Representation: The core idea of CSP-MDKernel is to enhance feature representation through feature partitioning and multidomain modeling. Specifically, the input feature map is partitioned into two parts at a ratio of 1:3. This partitioning strategy greatly reduces computational overhead. Only one quarter of the features is processed by the computationally intensive MDKernel [as shown in Fig. 3(b)] to capture representations from multiple scales and domains, while the remaining three quarters are passed through directly. This approach preserves feature integrity while avoiding excessive parameter growth. Within the MDKernel architecture, three branches—global, large-scale, and local—are introduced, each optimized for different feature extraction requirements.

1) The **global branch** [as shown in Fig. 3(c)] employs a frequency-domain channel attention (FDCA) module, a spatial-channel attention module (SCAM), and a frequency-guided module (FGM) to enhance the representational capacity of the input features progressively. Specifically, FDCA utilizes a 1×1 convolution to compute channel attention weights and employs the discrete wavelet transform (DWT) to decompose the input features into four frequency subbands: LL, LH, HL, and HH. Attention is applied exclusively to the LL subband, which encodes structurally stable and semantically meaningful information, thereby enhancing global representation robustness and suppressing background noise. In contrast, applying attention directly to HH components may amplify irrelevant textures or noise. The enhanced LL features are then recombined with the original HH subbands (LH, HL, and HH) through an inverse DWT (IDWT) to reconstruct the spatial feature map. This strategy strengthens LL structures while preserving raw HH details. SCAM extracts multiscale features through local and global pooling operations, followed by upsampling to restore the original resolution and generate a spatial-channel attention map. The attention map is weighted to the output of the FDCA module, reinforcing critical region representations and minimizing the impact of uninformative zones. FGM extracts frequency information using depthwise convolution and fast Fourier transform (FFT) and performs element-wise multiplication in the frequency domain to emphasize critical frequency components. The result is subsequently reverted to the spatial domain using an inverse FFT (IFFT) operation.

Although both the DCA module and the global branch of MDKernel utilize global contextual information, they serve distinct roles at different stages of the network. DCA, introduced in Section III-B, is embedded in the backbone and employs AGAP to guide directional convolutions along horizontal and vertical axes. This design enhances the model's sensitivity to weak contours and directional textures in small objects. In contrast, the global branch of MDKernel, positioned in the neck network (MDAP), focuses on high-level semantic enhancement by integrating global cues across the frequency, spatial, and channel domains. This design compensates for the semantic limitations of shallow high-resolution features. Collectively, DCA and the MDKernel global branch perform complementary, nonredundant roles within the detection pipeline.

2) The **large-scale branch** employs multiple convolutional kernels (31×31, 31×1, and 1×31) to extract global contextual information and horizontal and vertical directional features, respectively.

3) The **local branch** emphasizes local features through 1×1 convolutions, enhancing the perception of small objects and edge details.

2) SCFConv: Optimizing Feature Interaction and Diversity: SCFConv is designed to optimize the interaction of local

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

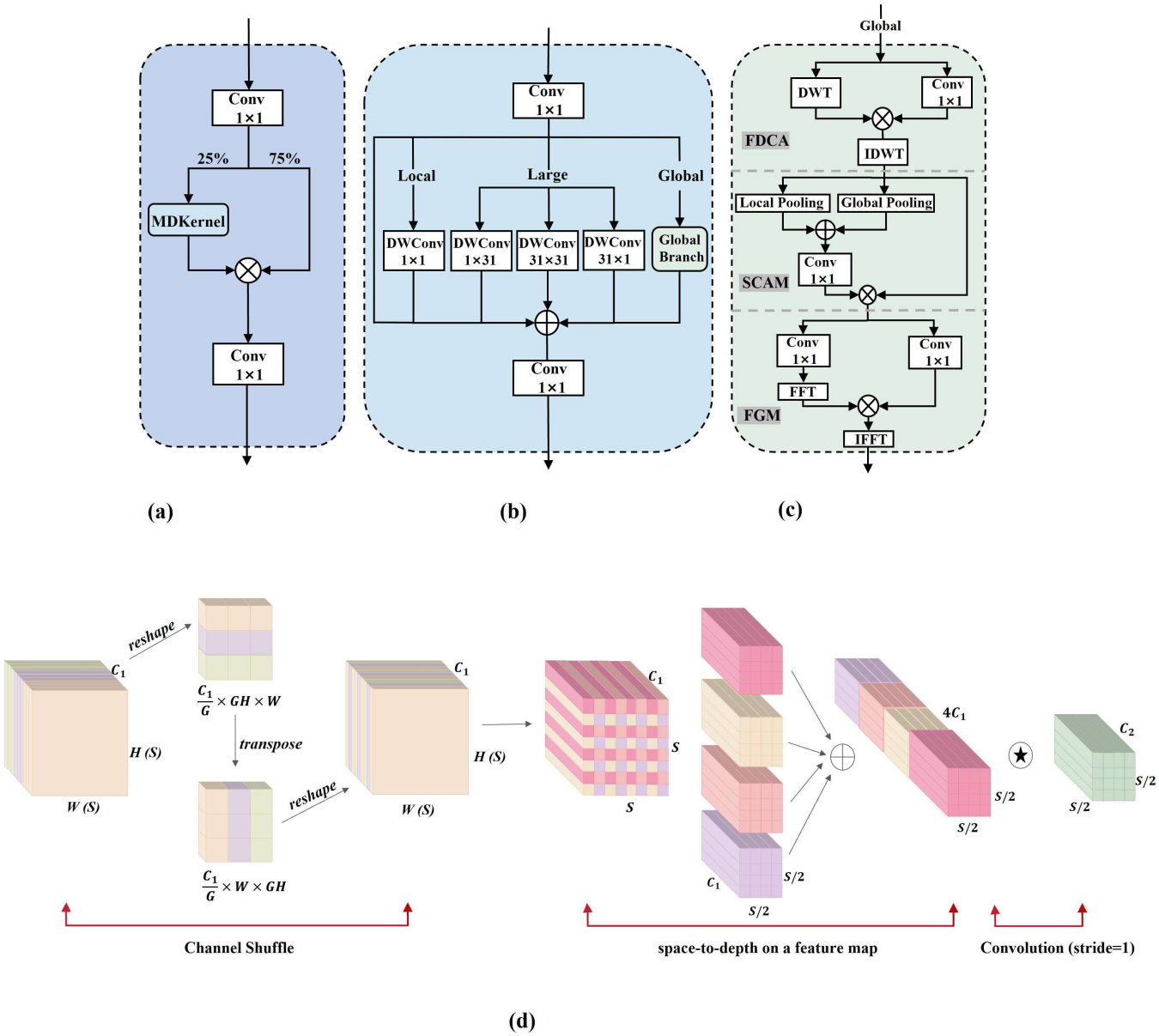


Fig. 3. Architecture of the MDAP. (a) CSP-MDKernel, which partitions the input feature map in a 1:3 ratio, selectively enhancing one-quarter through multidomain processing. (b) MDKernel with local, large-scale, and global branches. (c) Global branch combines FDCA, SCAM, and FGM to enhance feature representations across frequency, spatial, and channel domains. (d) SCFConv reorganizes features by spatial and channel partitioning, aggregates region-specific responses, and applies a 3 × 3 convolution to preserve continuity and enhance fine-grained object representation.

features and enhance feature diversity and representational capacity. This module consists of four key steps as follows.

Step 1 (Channel Shuffle): A channel reorganization strategy is employed, in which the input features are split into G groups along the channel dimension, with each group consisting of C/G separate channels (where C is the number of input channels, and G is the number of groups). Exchanging the positions of groups and channels enhances feature diversity.

Step 2 (Information-Preserving Spatial Decomposition): The feature map is decomposed into four subfeature maps using a checkerboard-style sampling strategy with a stride of 2. Specifically, for an input feature map X of spatial dimensions $H \times W$ and C channels, we extract pixels from nonoverlapping positions: the top-left submap from $X[\ldots, :: 2, :: 2]$, top-right from $X[\ldots, :: 2, 1 :: 2]$, bottom-left from $X[\ldots, 1 :: 2, :: 2]$, and bottom-right from $X[\ldots, 1 :: 2, 1 :: 2]$. This process reduces the spatial size of each submap to $H/2 \times W/2$ while preserving all pixel information through a spatial reorganization. This spatial partitioning is not intended to segment objects but serves as an intermediate step for fine-grained regional feature extraction. It increases spatial diversity and enables the network to focus more effectively on localized detail representations.

Step 3 (Channel Concatenation): The four subfeature maps from Step 2 are concatenated along the channel dimension, generating a fused representation that increases the channel dimension by a factor of 4. This operation effectively transfers

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

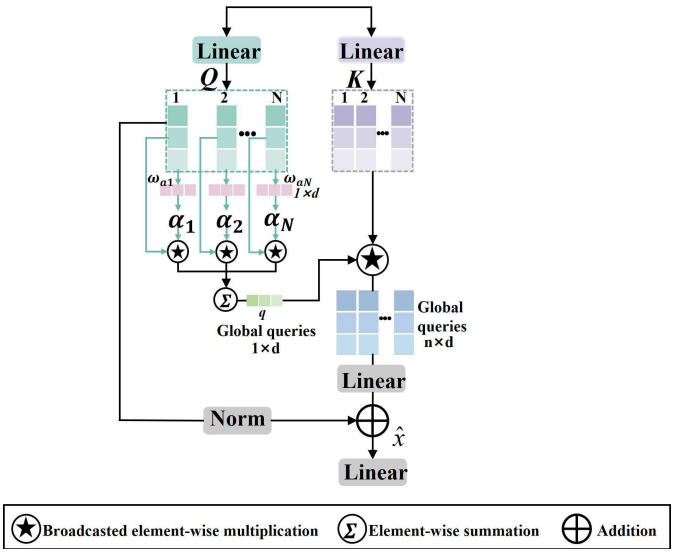


Fig. 4. Structure of the EfficientAIFI. The module replaces traditional QKV operations with a query–key interaction mechanism, enabling global modeling through linear computation.

the original spatial information into the channel dimension without loss.

Step 4 (Feature Transformation With Nonstrided Convolution): A nonstrided (stride = 1) 3 × 3 convolution is applied to the concatenated feature map. The primary role of this convolution is to integrate the features across the expanded channel dimension and project them down to a desired, smaller number of output channels. Using a stride of 1, this operation avoids the information loss associated with strided convolution, ensuring that all the information preserved by the SCFConv is utilized to learn a compact and effective feature representation.

In the MDAP structure, the high-resolution features of the P2 layer are preserved, thereby enhancing the detailed information of small objects. SCFConv further optimizes these features, enabling them to capture both local details and global contextual information when fused with the P3 layer. Meanwhile, CSP-MDKernel enhances the integration of features, further strengthening the representation of both global and local features and significantly improving the detection performance of small objects.

D. EfficientAIFI Module

In the intrascale interaction module, the MHSA mechanism serves as the core component, capturing global relationships within the input sequence by parallelizing the computation of multiple attention heads, thereby enhancing the model's expressive capability. However, MHSA suffers from high computational complexity ($O(N^2)$), resource wastage (e.g., uninformative attention connections), and significant overhead in multihead computations, especially in long-sequence scenarios where efficiency drastically decreases. To address these issues, this article proposes EfficientAIFI, which replaces the conventional MHSA with EAA (as illustrated in Fig. 4). Unlike traditional self-attention mechanisms, EAA discards the interaction between keys and values, retaining only the relationship between queries and keys. It replaces the costly matrix multiplication with efficient element-wise linear multiplication.

Specifically, the input tensor X is passed through two fully connected layers, namely, to_query and to_key, which project it into a query matrix Q and a key matrix K, respectively,

$$\begin{aligned} Q &= \text{normalize}\left(\text{to_query}\left(X\right)\right) \\ K &= \text{normalize}\left(\text{to_key}\left(X\right)\right) \end{aligned} \tag{5}$$

where the normalization operation ensures that the vector norms of the queries and keys are equal to 1.

Subsequently, the attention weight for each query is computed by taking the inner product between the query matrix Q and a learnable parameter w_∂. The attention weights A are calculated as follows:

$$A = \text{normalize}\left(Q \cdot w_\partial\right) \cdot \text{scale_factor} \tag{6}$$

where the scale factor is a constant (typically set to $1/\sqrt{d}$, where d denotes the dimensionality of the queries), which is used to mitigate the issue of excessively large query weights and to ensure numerical stability during training.

Subsequently, the global context G is obtained by performing a weighted aggregation over the queries

$$G = \sum_i A_i \cdot Q_i \tag{7}$$

where A_i denotes the attention weight of the ith query, and Q_i represents the corresponding query vector.

The global context G is replicated to match the dimensionality of the key matrix K, and an element-wise multiplication is performed to achieve efficient interaction between queries and keys

$$\begin{aligned} G_{\text{repeat}} &= \text{repeat}\left(G, \text{shape} = K_{\text{shape}}\right) \\ \text{Output} &= T\left(G_{\text{repeat}} \cdot K\right) + Q. \end{aligned} \tag{8}$$

Subsequently, a linear transformation T is applied to refine the feature representations further, ensuring the conveyed information's effectiveness.

Finally, the output tensor is generated through a linear transformation layer with a shape of $[B, N, D]$, where B denotes the batch size, N is the sequence length, and D is the feature dimension. A subsequent reshape operation restores it to the original 4-D tensor $[B, C, H, W]$, facilitating further processing.

E. Auxiliary BSG

In remote sensing imagery, cluttered backgrounds and fine textures frequently induce spurious activations near object boundaries, resulting in false detections even for advanced detectors. To mitigate this issue in a lightweight manner, we propose a lightweight BSG as an auxiliary module. BSG is designed to suppress background-induced noise and enhance the robustness of the detection framework in cluttered scenes.

Specifically, let $F \in \mathbb{R}^{C\times H\times W}$ denote the highest-resolution feature map from the detection pyramid. A 1 × 1 convolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

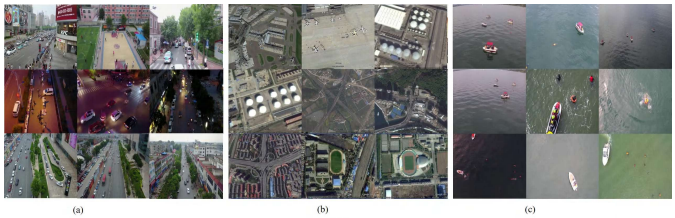


Fig. 5. Sample from (a) VisDrone-DET, (b) RSOD, and (c) SeaDronesSeeV2 datasets are shown.

is applied to compress channel information, followed by a Sigmoid activation to generate a normalized gate map

$$g = \sigma\left(W_{1\times1} * F\right) \tag{9}$$

where $W_{1\times1}$ denotes the convolution kernel and $\sigma(\cdot)$ is the Sigmoid function. The resulting gate map $g \in [0,1]^{H\times W}$ encodes a background-likeness prior, with higher values indicating stronger responses to background regions.

For each predicted bounding box, its normalized center coordinates (x, y) are projected onto the gate map, and the corresponding gate value $g(x, y)$ is obtained via bilinear interpolation. This operation ensures differentiability and spatial smoothness when sampling from the continuous gate map. Finally, let s denote the raw classification logit for a given prediction. The BSG refines this logit by applying a background-aware penalty

$$s' = s - \alpha \cdot g\,(x, y) \tag{10}$$

where α is a learnable scalar controlling the suppression strength. A larger gate response $g(x, y)$ results in a stronger reduction in s', effectively downweighting detections in background-dominated regions. This auxiliary design suppresses spurious activations caused by cluttered textures while preserving genuine object responses, complementing the three main modules to enhance overall robustness.

IV. Experiments

A. Datasets

VisDrone-DET is a benchmark dataset for object detection from UAV, released in 2018 by researchers from the AISKYEYE team affiliated with Tianjin University. The dataset comprises 10 209 high-resolution static images (960 × 540 to 2000 × 1500) and 288 video clips collected from diverse and complex scenes across 14 cities. It covers ten object categories, including pedestrians and vehicles. This dataset has been partitioned into three subsets: 6471 instances for training, 548 for validation, and 3190 for testing. Its significant class and scale imbalance makes it an ideal testbed for small object detection research. Example samples from the dataset are shown in Fig. 5(a).

RSOD is an open-source remote sensing image detection dataset consisting of 976 images sourced from Google Earth and TianDiTu, with sizes ranging from 512 × 512 to 1961 × 1193 pixels. It includes annotations for four object categories—aircraft, playgrounds, overpasses, and oil tanks—containing 6950 instances. The dataset is randomly partitioned into training, validation, and test subsets using a 6:2:2 split ratio, making it suitable for evaluating object detection algorithms in remote sensing scenarios. Example samples from the dataset are shown in Fig. 5(b).

SeaDronesSeeV2 is a large-scale maritime detection dataset designed for offshore rescue missions, containing 8930 training images and 1547 validation images, most of which drones capture at 4K resolution. The dataset covers five object categories: boat, swimmer, life_saving_appliances, jet ski, and buoy, providing a dedicated benchmark for small object detection and localization tasks in marine environments. Example samples from the dataset are shown in Fig. 5(c).

B. Experimental Environment Setup

The experimental settings of this study are as follows: an NVIDIA GeForce RTX 3090 Ti GPU was used, with CUDA version 11.7, Python 3.8.13, and PyTorch 1.12.1. The optimizer was AdamW. The batch size was configured as 8 with a momentum parameter of 0.9. The training was conducted for 350 epochs, using an initial learning rate of 1×10^{-4} and applying a weight decay parameter of 1×10^{-4}.

C. Evaluation Indicators

This study employs multiple evaluation metrics to comprehensively assess model performance, including recall (R) [see (11)], precision (P) [see (12)], average precision (AP) [see (13)], floating-point operations (GFLOPs), and number of model parameters (Params), as well as AP scores for objects of different scales (APs, APm, and APl). In addition, mean AP (mAP) is used to evaluate overall accuracy, while AP50, AP75, and AP50–95 further measure the model's performance at specific intersection over union (IoU) thresholds

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{11}$$

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{12}$$

$$\mathrm{AP} = \int_0^1 P\,(R)\,dR \tag{13}$$

where AP is derived from the precision–recall curve, where precision and recall are computed based on FP, TP, and FN. It represents the area under this curve.

mAP@0.5 represents the mAP calculated for all categories at an IOU threshold of 0.5. At the same time, mAP@0.5:0.95 denotes the average of AP values computed at multiple IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05. AP50 and AP75 refer to the AP at IoU thresholds of 0.5 and 0.75, respectively, and AP50–95 is the mAP over the IoU range from 0.5 to 0.95 (step size 0.05). APs evaluates the AP for small objects (typically with an area less than 32 × 32 pixels), APm assesses medium-sized objects (with an area between 32 × 32 and 96 × 96 pixels), and APl measures large objects (with an area greater than 96 × 96 pixels).

In addition, GFLOPs measure the floating-point computations required for inference, reflecting the model complexity. Params indicate the number of parameters the model contains, serving as a measure of its storage demand and structural

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARATIVE EXPERIMENTS ON THE RSOD DATASET

Method	Publication	mAP@0.5 (%)	Airplane (%)	Oil-tank (%)	Overpass (%)	Playground (%)
Sig-NMS [44]	IEEE TGRS 2019	89.4	80.6	90.6	87.4	99.1
Cascade R-CNN [4]	CVPR 2018	91.3	94.2	96.1	83.2	99.0
EfficientDet [9]	CVPR 2020	82.8	85.7	92.6	54.6	98.2
CANet [45]	IEEE TGRS 2021	95.2	91.7	97.0	**94.1**	97.9
YOLOv8	-	95.7	96.8	98.1	88.6	99.3
YOLOv11n	-	92.0	-	-	-	-
SCDFMixer [43]	IEEE Sens. J. 2024	**96.2**	**97.8**	**98.1**	89.4	99.5
SMA-YOLO [46]	SR 2025	95.4	-	-	-	-
SAPF [47]	IEEE TGRS 2025	90.4	-	-	-	-
DETR [13]	ECCV 2020	89.6	92.3	88.4	85.5	93.2
RT-DETR-r18 [14]	CVPR 2024	94.8	93.1	97.1	89.4	**99.5**
Ours	-	**97.8**	**98.1**	**99.2**	**93.9**	**100.0**

TABLE II
COMPARATIVE EXPERIMENTS ON THE VISDRONE-DET DATASET

Method	Publication	APs (%)	APm (%)	APl (%)	AP50-95 (%)	AP50 (%)	AP75 (%)	GFLOPS
Cascade R-CNN [4]	CVPR 2018	15.2	36.7	39.2	24.5	39.0	26.1	120.3
RetinaNet [10]	ICCV 2017	13.7	37.4	41.0	23.5	40.2	23.8	-
CEASC [50]	CVPR 2023	-	-	-	28.7	**50.7**	28.4	150.2
ClusDet [51]	ICCV 2019	-	-	-	26.7	50.6	24.7	-
MSFE-YOLO-m [52]	IEEE GRSL 2024	16.1	39.4	-	27.5	44.5	-	-
SDMNet [53]	Image Vision Comput. 2024	-	-	-	27.5	46.9	-	378.3
Cascade ADPN [54]	Neurocomputing 2022	12.3	33.8	41.8	22.8	38.7	23.8	547.2
αS-YOLO [48]	IEEE J-STARS 2025	17.2	**41.3**	**62.8**	**30.4**	48.3	31.6	-
QueryDet [55]	CVPR 2022	19.8	35.9	40.3	28.3	48.1	28.8	212.0
DINO [49]	ICCV 2021	**22.9**	38.4	44.5	29.4	46.2	**29.1**	279.0
UAV-DETR [56]	arXiv2025	-	-	-	29.8	48.8	-	**77.0**
DMR-RTDETR [57]	IEEE TIM	19.5	-	-	27.9	46.0	28.7	-
RT-DETR-r18 [14]	CVPR 2024	18.2	35.6	41.0	26.2	44.0	26.4	**57.0**
Ours	-	**23.0**	**42.0**	**47.8**	**30.7**	**50.8**	**31.6**	83.5

TABLE III
COMPARATIVE EXPERIMENTS ON THE SEADRONESSEEV2 DATASET

Method	Publication	Backbone	AP50-95 (%)	AP50 (%)	AP75 (%)
Faster-RCNN [4]	NeurIPS 2015	ResNet-R50	31.3	61.7	28.9
FCOS	ICCV 2019	ResNet-R50	42.5	71.6	47.2
YOLOv5 [60]	-	CSPDarkNet	42.6	77.0	40.9
TPH-YOLOv5 [61]	ICCV 2021	CSPDarkNet	46.4	81.1	47.4
FFDFA-YOLO [62]	ISCIPT 2024	CSPDarkNet	50.1	78.8	-
YOLOv10-n [58]	-	CSPDarkNet	33.6	58.9	33.7
YOLOv8	-	CSPDarkNet	41.7	68.0	43.4
YOLOv7-tiny [63]	CVPR 2023	CSPDarkNet	40.2	72.9	38.9
EUAVDet [64]	Drones 2024	-	41.3	76.7	39.3
F.ResNeXt101-FPN-FL [65]	SR 2024	-	36.3	68.8	30.0
MFDS-DETR [59]	CIBM 2024	ResNet-R50	40.6	79.0	43.3
OWRT-DETR [15]	IEEE TGRS2025	LFENet	**50.3**	**84.9**	47.9
RT-DETR-r18 [14]	CVPR 2024	ResNet-R18	49.7	83.2	**49.1**
Ours	-	SAAB	**50.7**	**84.2**	**52.0**

complexity. Elevated parameters may enhance learning ability but potentially lead to overfitting.

D. Quantitative Analysis

In this section, the detection performance of SME-DETR is compared with several existing models on the RSOD, VisDrone-DET, and SeaDronesSeeV2 datasets to validate its effectiveness and advantages. It is important to note that all comparative results in this section are obtained using the core SME-DETR framework without the auxiliary BSG module, ensuring a fair comparison with baseline methods. The comparison models include CNN-based models and transformer-based models. The detailed comparative results are presented in Tables I–III.

1) Comparison on the RSOD Dataset: To assess SME-DETR's performance, a comparison with state-of-the-art approaches was conducted using the RSOD dataset, as summarized in Table I. For clarity, the best results for each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION EXPERIMENTS ON THE VISDRONE-DET DATASET

Visdrone-val	SAA Backbone	EfficientAIFI	MDAP	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)	APs (%)	Params (M)	GFLOPs
A (baseline)				61.6	45.6	47.0	28.7	18.2	19.88	57.0
B	√			63.2	48.7	50.0	30.4	19.9	**17.24**	**54.2**
C		√		62.0	47.4	48.8	29.6	19.5	19.88	57.2
D			√	62.6	49.0	49.7	30.6	19.9	20.50	65.4
E	√	√		64.5	50.8	52.0	32.3	22.6	17.24	54.5
F		√	√	64.2	48.6	50.4	31.0	20.5	20.50	65.7
G	√	√	√	**65.2**	**52.2**	**53.0**	**33.1**	**23.0**	19.47	83.5

TABLE V
ABLATION EXPERIMENTS ON THE RSOD DATASET

RSOD	SAA Backbone	EfficientAIFI	MDAP	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)	APs (%)	Params (M)	GFLOPs
A (baseline)				93.8	93.1	94.8	64.2	39.2	19.88	57.0
B	√			94.0	94.5	95.3	66.2	42.9	**17.24**	**54.2**
C		√		93.1	92.7	95.3	64.8	42.6	19.88	57.2
D			√	88.1	92.2	95.6	66.0	**44.5**	20.49	65.4
E	√	√		94.0	92.9	95.7	67.3	43.4	17.24	54.4
F		√	√	**94.6**	92.5	97.1	66.6	43.9	20.49	65.6
G	√	√	√	93.2	**94.5**	**97.8**	**68.2**	44.0	19.46	83.4

category are shown in bold, and the second-best is marked in bold green. The experimental results indicate that our model achieves 100.0% precision in detecting the playground category, representing a 0.5% improvement over RT-DETR and a 4.5% increase in the overpass category. Moreover, SME-DETR achieved the highest mAP@0.5 score of 97.8%, surpassing the second-best SCDFMixer [43] by 1.6%, DETR by 8.2%, and the baseline RT-DETR by 3.0%. This significant improvement demonstrates that our approach surpasses most competing approaches on the RSOD dataset, showcasing outstanding performance in this domain.

2) Comparison on the VisDrone-DET Dataset: To assess SME-DETR's performance, a comparison with state-of-the-art approaches was conducted using the VisDrone-DET dataset, as summarized in Table II. As with the RSOD dataset, the best result in the table is shown in bold, while the second-best is marked in bold green. The experimental results indicate that our model achieves outstanding performance across multiple key metrics, particularly in APs, APm, AP50–95, AP50, and AP75, significantly surpassing existing methods such as RT-DETR-r18. For example, SME-DETR achieves an APs of 23.0% for small object detection, considerably higher than the 18.2% achieved by RT-DETR-r18; for medium object detection (APm), SME-DETR reaches 42.0%, exceeding RT-DETR-r18 by 6.4%. In terms of the AP50 metric, SME-DETR achieves 50.8%, outperforming αS-YOLO [48] (48.3%) and DINO [49] (46.2%); for AP75, SME-DETR attains 31.6%, representing a 5.2% improvement over RT-DETR-r18 (26.4%). These results demonstrate that the proposed method exhibits stronger generalization ability and stability across object detection tasks at different scales.

3) Comparison on the SeaDronesSeeV2 Dataset: To assess SME-DETR's performance, a comparison with state-of-the-art approaches was conducted using the SeaDronesSeeV2 dataset, as shown in Table III. Following the same convention as in the previous datasets, the best result is shown in bold, and the second-best is marked in bold green. The experimental results indicate that SME-DETR outperforms existing methods across key metrics such as AP50–95, AP50, and AP75. For instance, SME-DETR achieves an AP50–95 of 50.7%, which clearly exceeds the performance of RT-DETR-r18 (49.7%) and significantly surpasses YOLOv10-n [58] (33.6%) and MFDS-DETR [59] (40.6%). Regarding AP50, SME-DETR reaches 84.2%, outperforming RT-DETR-r18 (83.2%) and MFDS-DETR (79.0%). Moreover, SME-DETR attains an AP75 of 52.0%, representing a 2.9% improvement over RT-DETR-r18 (49.1%), demonstrating superior detection capabilities under high IoU thresholds.

E. Ablation Experiments

This section presents six ablation studies to evaluate the effectiveness of the core modules in the SME-DETR framework. To clearly isolate the contributions of the core components, the ablation studies in Tables IV–X are conducted without the auxiliary BSG module. The effect of BSG as an auxiliary enhancement is separately analyzed in Table XI. All experiments are conducted under consistent settings on two UAV-based object detection datasets: VisDrone-DET and RSOD. The corresponding results are summarized in Tables IV–XI.

The step-by-step introduction of each module leads to varying degrees of performance improvement, as summarized in Tables IV and V. Experiment A serves as the baseline for comparison. Replacing the baseline ResNet18 with the proposed SAA-Backbone significantly improves mAP@0.5:0.95 and mAP@0.5 while lowering the number of Params and GFLOPs, particularly on the VisDrone-DET dataset, due to the DPF design that improves multiscale feature extraction. Substituting

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION STUDY OF SUBMODULES IN THE SAA-BACKBONE (VISDRONE-DET)

Visdrone-val (dataset)	Model / Module Combination	mAP@0.5 (%)	mAP@0.5:0.95 (%)	APs (%)	Params (M)	GFLOPs
A	Baseline (ResNet18)	47.0	28.7	18.2	19.88	57.0
B	+ DCA	48.9	29.3	19.5	17.24	**47.8**
C	+ DSA	49.5	29.7	19.6	17.24	54.2
D	SAA	**50.0**	**30.4**	**19.9**	**17.24**	54.2

TABLE VII
ABLATION STUDY ON THE EFFECT OF KERNEL SIZE *n* IN THE DCA MODULE (VISDRONE-DET)

Visdrone-val (dataset)	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)	APs (%)	Params (M)	GFLOPs
Baseline	61.6	45.6	47.0	28.7	18.2	19.88	57.0
$n = 0$	62.3	46.9	48.4	28.5	18.7	17.24	54.2
$n = 1$	62.5	48.1	49.5	29.7	**20.4**	17.24	54.2
$n = 2$	62.8	48.4	49.7	29.8	20.1	17.24	54.2
$n = 3$	**63.2**	**48.7**	**50.0**	**30.4**	19.9	**17.24**	**54.2**
$n = 4$	62.6	48.0	49.0	28.9	19.4	17.24	54.2
$n = 5$	62.8	47.5	49.0	30.0	19.8	17.24	54.2

TABLE VIII
ABLATION EXPERIMENTS ON DIFFERENT FEATURE PARTITION RATIOS IN CSP-MDKERNEL (VISDRONE-DET)

Ratio	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)	APs (%)	Params (M)	GFLOPs
1:1	62.3	47.9	48.9	29.9	19.3	20.8	68.8
1:2	62.2	47.1	48.3	29.6	19.6	20.6	66.4
1:3	**62.6**	**49.0**	**49.7**	**30.6**	**19.9**	**20.5**	65.4
1:4	61.7	46.5	47.9	29.4	18.9	20.5	**64.9**

TABLE IX
ABLATION STUDY ON SCFCONV CONFIGURATIONS (VISDRONE-DET)

Method	mAP@0.5 (%)	mAP@0.5:0.95 (%)	APs (%)	Params (M)	GFLOPs
Baseline	47.0	28.7	18.2	**19.88**	**57.0**
SCFConv (No Fusion)	47.2	28.8	18.0	19.88	57.5
SCFConv (Full)	**47.6**	**29.2**	**19.3**	21.03	61.2

TABLE X
ABLATION STUDY OF DOMAIN-SPECIFIC MODULES IN THE MDAP STRUCTURE (RSOD)

RSOD (dataset)	FDCA	SCAM	FGM	mAP@0.5 (%)	mAP@0.5:0.95 (%)	APs (%)
A	√			93.7	63.8	39.1
B		√		92.9	64.4	37.6
C			√	92.8	64.1	39.6
D	√	√		94.2	65.2	43.5
E (Full)	√	√	√	**95.6**	**66.0**	**44.5**

the original AIFI with EfficientAIFI boosts mAP@0.5:0.95, mAP@0.5 and APs, with negligible changes to Params and GFLOPs, owing to the efficient attention mechanism that reduces noise while preserving global context. Introducing the MDAP structure on the RSOD dataset achieves the highest APs performance by leveraging multidomain fusion and preserving fine-grained details. Integrating the SAA-Backbone with the EfficientAIFI module offsets the parameter and computational overhead of using EfficientAIFI alone and further improves detection accuracy. Combining EfficientAIFI and MDAP leads to enhancements in key performance indicators such as precision, mAP@0.5:0.95, and mAP@0.5. Finally, the detection performance is significantly enhanced when all three modules are integrated (Experiment G). On the VisDrone-DET dataset, mAP@0.5:0.95 and mAP@0.5 increase by 6.0% and 4.4%, respectively, while APs improve by 4.8% compared

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
EVALUATION OF THE AUXILIARY BSG (RSOD)

Method	AP50-95 (%)	AP50 (%)	AP75 (%)	Params (M)	GFLOPs	FP $\triangle$AP $\downarrow$
RT-DETR-r18	63.5	93.5	68.9	19.88	57.0	3.77
Ours w/o BSG	**67.1**	94.1	76.3	19.46	**83.4**	4.14
Ours + BSG	67.0	**96.1**	**77.1**	**19.46**	84.3	**1.73**

to the baseline. On the RSOD dataset, mAP@0.5:0.95 and mAP@0.5 increase by 3.0% and 4.0%, respectively.

These results validate the effectiveness of each module and demonstrate their complementary strengths when combined, highlighting the overall robustness and scalability of the SME-DETR framework in diverse UAV-based remote sensing scenarios.

To evaluate the individual and combined effects of the DCA and DSA modules, we conducted an ablation study using four model variants: baseline (ResNet18), +DCA, +DSA, and the full SAA configuration. As shown in Table VI, both DCA and DSA improve detection performance over the baseline, with DSA yielding slightly higher gains in mAP@0.5:0.95, mAP@0.5, and APs. Notably, both modules also reduce the number of parameters and GFLOPs. The full SAA configuration achieves the best results—mAP@0.5:0.95 of 30.4%, mAP@0.5 of 50.0%, and APs of 19.9%—demonstrating the complementary benefits of scale-aware and direction-aware feature modeling.

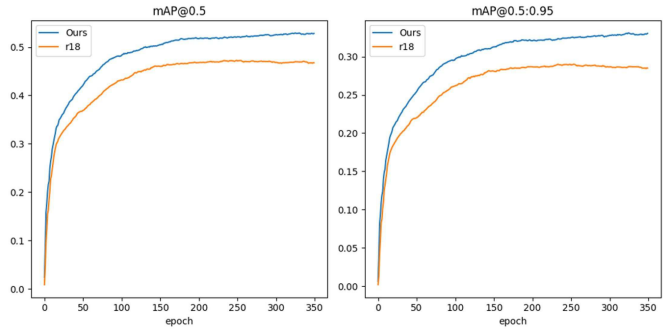
To evaluate the influence of horizontal/vertical convolution kernel size on edge modeling, we conduct an ablation study by varying the kernel size parameter $n \in \{0, 1, 2, 3, 4, 5\}$ in the DCA module. Each value of n corresponds to convolutional filters of size $(1, 2n + 1)$ and $(2n + 1, 1)$ for horizontal and vertical branches, respectively. As shown in Table VII, increasing n from 0 to 3 steadily improves mAP@0.5 and mAP@0.5:0.95, indicating stronger edge modeling and spatial context awareness. However, when n exceeds 3, the performance tends to plateau or slightly decrease due to oversmoothing and noise interference from larger receptive fields. Notably, while the APs for $n = 1$ are slightly higher than that for $n = 3$, the latter provides the best overall performance. Small kernel sizes may preserve more fine-grained details but fail to capture sufficient global context. Therefore, $n = 3$ is adopted as the default configuration in our design. When applied directly to the RSOD and SeaDronesSeeV2 datasets, this setting yielded stable improvements, demonstrating robust transferability across diverse object distributions.

To further investigate the internal design of the MDAP module, we conduct an additional ablation study on the feature partitioning ratio in the CSP-MDKernel. Specifically, we compare the detection performance under four partition ratios: 1:1, 1:2, 1:3 (used in our default design), and 1:4, where the first part of the feature is processed by the MDKernel and the second is bypassed. As shown in Table VIII, the 1:3 configuration achieves the highest mAP@0.5 (49.7%), mAP@0.5:0.95 (30.6%), and APs (19.9%), while maintaining moderate GFLOPs (65.4). In contrast, despite involving more features, the 1:1 and 1:2 configurations result in lower performance, likely due to excessive redundancy or noise amplification in the complex branches of MDKernel. The 1:4 setting, though computationally lighter, suffers from limited domain-specific processing and degraded performance. These findings empirically confirm that the 1:3 ratio provides the optimal balance between accuracy and efficiency in CSP-MDKernel. When evaluated on RSOD and SeaDronesSeeV2, this same configuration sustained substantial performance gains, confirming its generalizability without dataset-specific tuning.

To evaluate the SCFConv design and examine potential feature fragmentation, we conduct a controlled ablation study under an identical network architecture. This setup ensures a fair comparison among three configurations: the baseline (RT-DETR-r18), SCFConv (no fusion), which applies spatial division (Step 2) but omits the subsequent channel concatenation and feature transformation (Steps 3 and 4), and the complete SCFConv (full) module. All compared variants utilize the same subsequent upsampling operations for dimensional consistency with the baseline FPN. As shown in Table IX, the no fusion variant yields only a marginal 0.2% gain in mAP@0.5 but a 0.2% drop in APs. This suggests that spatial partitioning alone provides negligible improvement and even impairs feature representation for small targets. The observed degradation highlights that partitioning is insufficient to preserve discriminative features, even with compensatory upsampling. In contrast, the complete SCFConv module yields consistent improvements across key detection metrics, achieving a mAP@0.5 of 47.6% (+0.6%) and an APs of 19.3% (+1.1%). These results confirm that the channel concatenation (Step 3) followed by the nonstrided 3×3 convolution (Step 4) effectively reintegrates the spatially partitioned features. This design mitigates potential fragmentation and enhances overall detection performance, particularly improving small-object detection by maintaining feature integrity across divided regions.

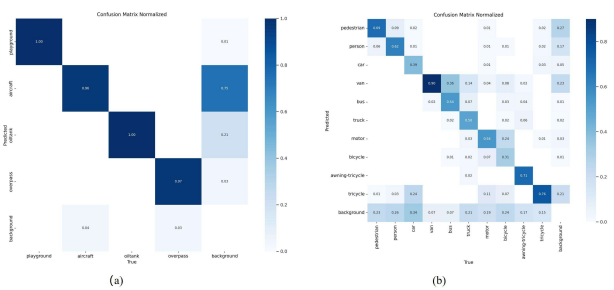
In addition, we further validate the individual contributions of each domain-specific module—namely, FDCA, SCAM, and FGM—within the MDAP structure through an ablation study on the RSOD dataset. As shown in Table X, we progressively activate and combine the three components to assess their respective impacts. Using FDCA alone (Experiment A) yielded a mAP@0.5 of 93.7% and APs of 39.1%, demonstrating its effectiveness in enhancing structural features via LL semantics. SCAM alone (Experiment B) achieved APs of 37.6% by focusing on spatially salient regions. FGM alone (Experiment C) achieved 39.6% APs by enhancing frequency-domain representations through FFT-based feature refinement. The combination of FDCA and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 6. Comparison of SME-DETR and RT-DETR-r18 on the VisDrone-DET dataset in terms of (a) mAP@0.5 and (b) mAP@0.5:0.95.

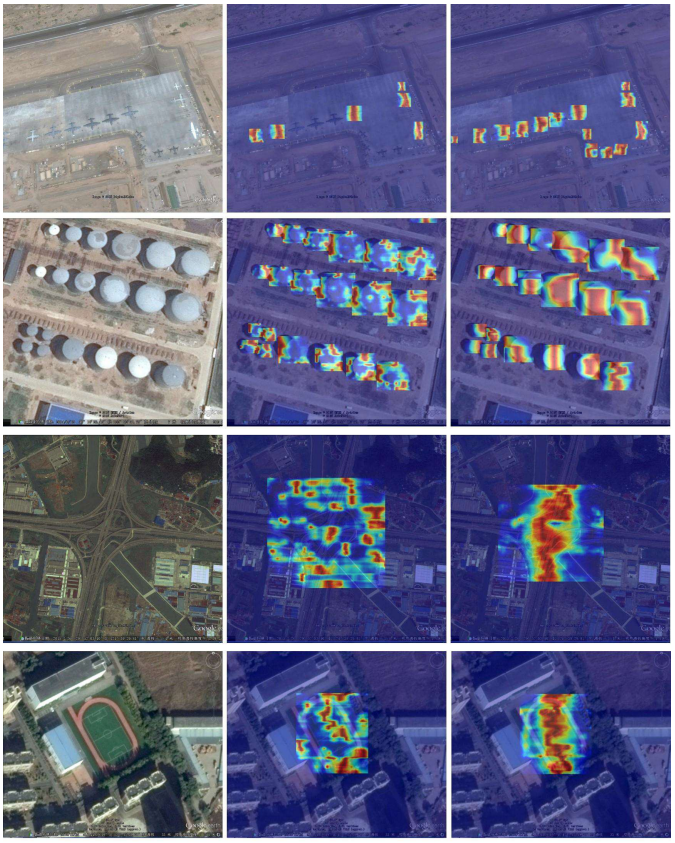


(a) (b)

Fig. 7. Confusion matrix results for (a) RSOD and (b) VisDrone-DET datasets.

SCAM (Experiment D) further improved APs to 43.5%, benefiting from joint modeling of global structure and spatial importance. Finally, the full MDKernel configuration (Experiment E), integrating all three modules, achieves the best overall performance with a mAP@0.5 of 95.6%, mAP@0.5:0.95 of 66.0%, and APs of 44.5%. These results demonstrate the complementary contributions of the domain-specific components and the effectiveness of the MDAP's multidomain fusion strategy in enhancing small object detection performance.

In addition to the ablation studies of the core modules, we further evaluate the auxiliary BSG. Remote sensing images frequently contain cluttered textures and fine-grained details that may induce spurious activations near object boundaries. To address this issue, BSG is incorporated as a lightweight enhancement that penalizes predictions influenced by background responses. As shown in Table XI, adding BSG to the core framework provides consistent improvements: AP50 increases from 94.1% to 96.1%, AP75 from 76.3% to 77.1%, while AP50–95 remains stable. The additional cost is negligible, with GFLOPs rising slightly from 83.4 to 84.3 and parameter counts unchanged. More importantly, false positives are substantially reduced, as indicated by FP ΔAP, which measures the reduction in AP caused by false positive detections. This value decreases markedly from 4.14 to 1.73, demonstrating fewer false positives. These results confirm that BSG effectively mitigates background-induced errors, enhancing the framework's robustness in cluttered scenarios without compromising efficiency.



(a) (b) (c)

Fig. 8. Comparison of heat map visualizations on the RSOD dataset. The second column of the image is the original model (RT-DETR-r18), and the third column is the SME-DETR model. (a) Original image. (b) RT-DETR-r18. (c) Ours.

F. Qualitative Analysis

1) Training Behavior and Classification Performance: The performance of SME-DETR and the baseline model RT-DETR-r18 was comparatively analyzed based on training accuracy and loss curves, as shown in Fig. 6. Over 350 training epochs: (a) and (b) illustrate the trends of mAP@0.5 and mAP@0.5:0.95, respectively, where SME-DETR (blue curve) exhibits a faster and more stable upward trend, indicating higher accuracy and better stability compared to RT-DETR-r18. In addition, the confusion matrix results (see Fig. 7) demonstrate that SME-DETR performs better in classification tasks.

2) Comparison of Heat Map Visualizations: As shown in Fig. 8, we visualize and compare the heatmaps produced by the baseline RT-DETR-r18 and the proposed SME-DETR. In these visualizations, the heatmap intensity reflects the model's focus or confidence in detecting objects—where warm colors (e.g., red and yellow) indicate higher attention weights. Compared with RT-DETR-r18, SME-DETR exhibits more concentrated and accurate local responses. This demonstrates the model's ability to effectively suppress background interference and allocate more discriminative attention to object regions, enhancing detection robustness. In several scenarios where RT-DETR-r18 fails to detect objects, SME-DETR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

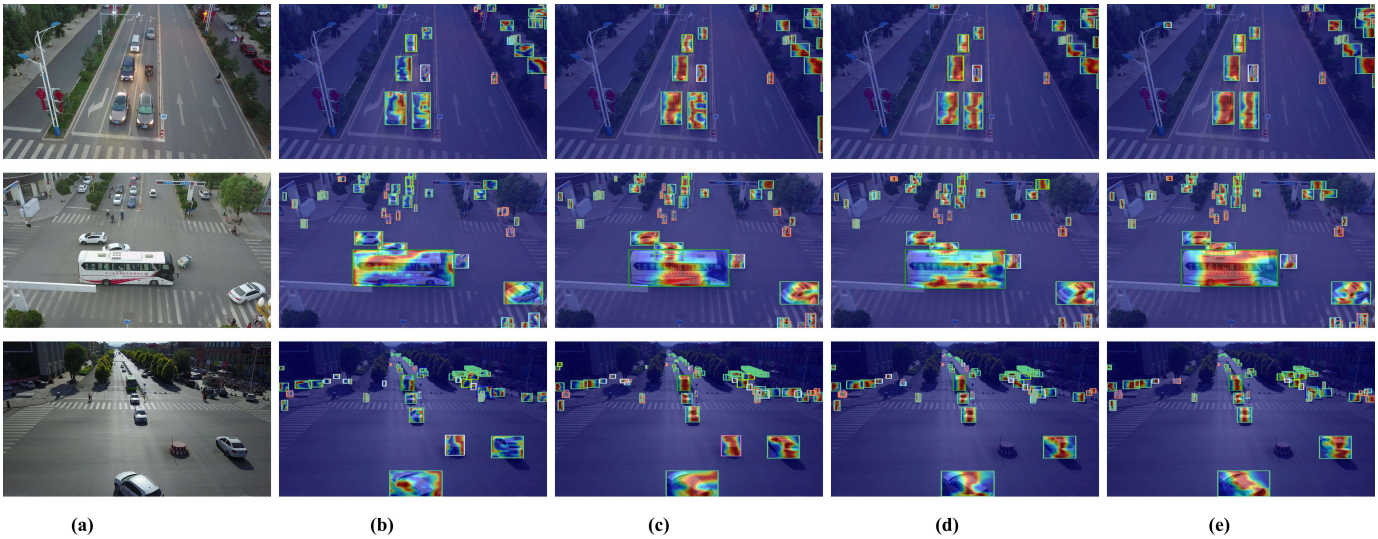


Fig. 9. LayerCAM attention map visualizations under different model configurations on VisDrone-DET. From left to right: (a) original image, (b) baseline, (c) +DCA, (d) +DSA, and (e) SAA.

successfully identifies them, as evidenced by the prominently activated areas in the heatmaps.

In addition, to further analyze the respective contributions of the DCA and DSA modules within the proposed SAA-Backbone, we visualize LayerCAM attention maps for four model variants—baseline, +DCA, +DSA, and the full SAA—on representative scenes from the VisDrone-DET dataset, as illustrated in Fig. 9. In the first two rows, the baseline model exhibits dispersed or misplaced activations, often failing to focus on fine-grained structures. In contrast, the +DCA variant improves edge alignment by enhancing horizontal and vertical boundary responses, while the +DSA variant provides more continuous and coherent texture activations for small objects. The full SAA configuration generates the most concentrated attention maps, effectively capturing both object boundaries and internal texture details. Notably, in the third row, both the baseline and +DCA variants mistakenly identify the traffic command platform as a van due to its strong edge contrast. In comparison, the +DSA and SAA variants successfully avoid this false detection, benefiting from their ability to model scale variation and spatial context. This case highlights the synergistic effect of direction-aware and scale-aware mechanisms, which jointly enhance boundary perception and suppress background interference, thereby improving detection accuracy and robustness.

3) Visualization of Improvement Effects: Visualization-based evaluations were carried out to demonstrate the improvement over the baseline RT-DETR-r18. The enhancements of SME-DETR on three datasets are shown in Fig. 10(a)–(c) (missed detections are highlighted using white circles, while false detections are outlined in blue boxes).

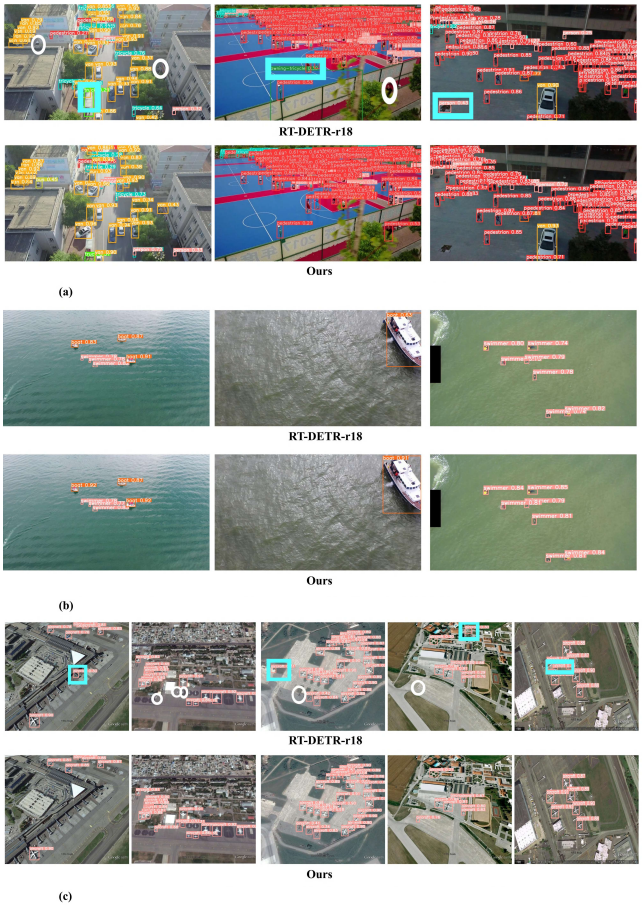


Fig. 10. (a)–(c) Comparison of RT-DETR-r18 and SME-DETR on the VisDrone-DET, SeaDronesSeeV2, and RSOD datasets.

In Fig. 10(a), the baseline model exhibits missed detections of small objects, such as buses, vans, and pedestrians, while SME-DETR accurately detects these objects. In addition, for the false detections of the truck, awning-tricycle, and person by the baseline model, SME-DETR successfully reduces these misclassifications. In Fig. 10(b), SME-DETR significantly improves detection accuracy over the baseline model, further enhancing its object recognition capability. In Fig. 10(c), SME-DETR also addresses the missed and false detections of the baseline model, such as more accurate detection of the aircraft category, further validating its superiority.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

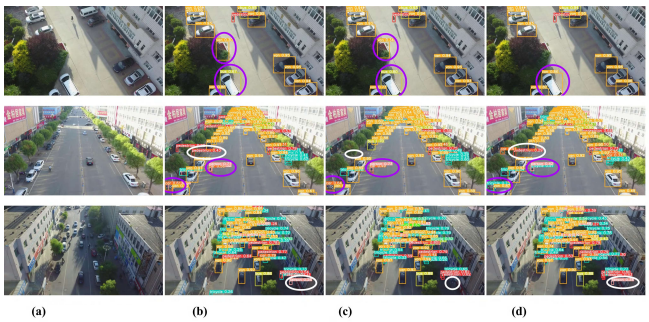


Fig. 11. Qualitative comparison of detection results on the VisDrone-DET dataset. Purple and white circles indicate false positives and missed detections, respectively. SCFConv (no fusion) produces false positives and missed detections in complex scenes, whereas SCFConv (full) accurately detects small and partially occluded targets. (a) Original image. (b) RT-DETR-r18. (c) SCFConv (no fusion). (d) SCFConv (full).

Overall, SME-DETR fully leverages the collaborative optimization of SAA-Backbone and MDAP, combined with EfficientAIFI, significantly improving the model's performance in small object detection. It effectively reduces missed and false detections, thereby enhancing the reliability and precision of object recognition.

4) Visual Analysis of SCFConv in Mitigating Feature Fragmentation: To evaluate the impact of spatial partitioning on feature integrity, we qualitatively compare the detection results on the VisDrone-DET dataset, as illustrated in Fig. 11. The results show that an intermediate variant, SCFConv (no fusion)—which executes spatial decomposition (Step 2) but omits the critical channel concatenation and feature transformation (Steps 3 and 4)—exhibits certain limitations compared with both the baseline detector and the complete SCFConv module.

As illustrated in the first row of Fig. 11, this variant not only inherits false positives from the baseline (e.g., misclassifying a partially occluded rectangular flowerbed as a van) but also produces higher confidence scores for these erroneous predictions. This indicates that the spatially partitioned feature representations weaken semantic coherence, thereby increasing the model's vulnerability to background interference. Furthermore, in the second and third rows, the no fusion variant fails to detect occluded or very small persons that are reliably identified by both the baseline detector and the complete SCFConv module. These observations collectively indicate that spatial partitioning alone compromises feature continuity and leads to fragmented representations of small or partially occluded objects.

Crucially, the complete SCFConv (full) module, which incorporates channel concatenation (Step 3) and the nonstrided 3×3 convolution for feature transformation (Step 4), effectively alleviates these limitations. As shown in the rightmost column of Fig. 11, the full module suppresses characteristic false positives and recovers missed detections observed in the no fusion variant. This demonstrates that the proposed fusion mechanism is essential for re-establishing cross-region dependencies and restoring the structural integrity of objects. The superior performance of the complete SCFConv confirms that the proposed design not only mitigates the feature-splitting issue but also establishes a more robust and coherent representation, which is critical for accurate detection in complex UAV-based imagery.

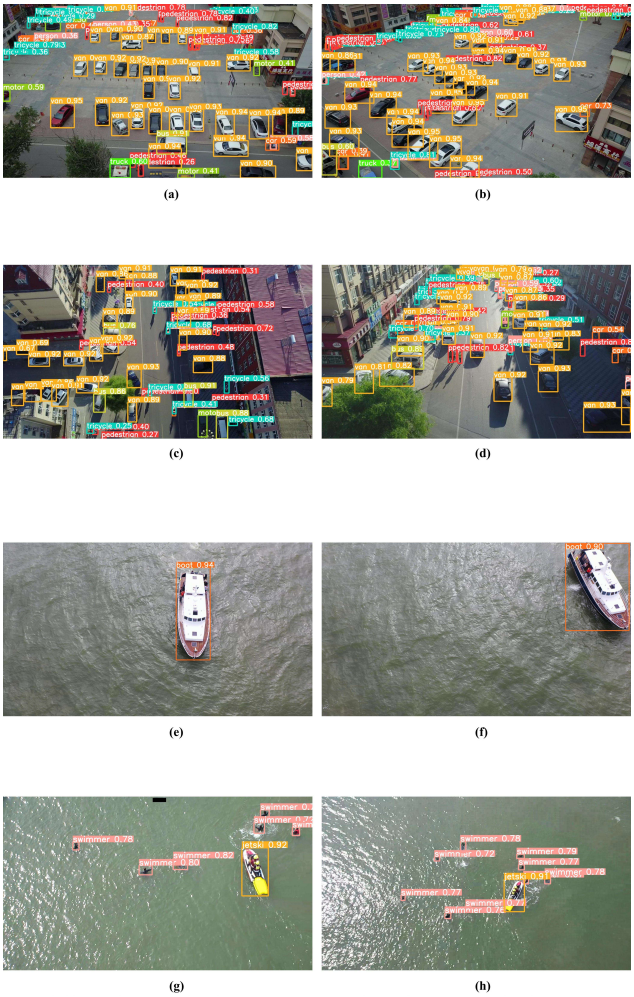


Fig. 12. Detection results under varying UAV viewpoints. The predicted bounding boxes remain accurate, with only slight fluctuations in confidence scores, demonstrating robust localization stability under moderate viewpoint changes. (a)-(b) Detection results on the same scene from the VisDrone-DET dataset under different UAV shooting angles. Despite the viewpoint change from a near-vertical (a) to an oblique angle (b), vans are consistently detected. (c)-(d) Detection results on the same urban street scene from the VisDrone-DET dataset under different UAV viewpoints. Although the UAV's orientation and altitude vary, the bounding boxes remain precise and well-aligned. While the confidence scores of some targets (e.g., vans and buses) slightly fluctuate, the overall detection stability is not significantly affected. (e)-(f) Detection results under different UAV viewing angles in the SeaDronesSeeV2 dataset. Despite the change from a vertical top-down view (e) to a more oblique side view (f), the boat is still accurately detected, with only slight variation in confidence score. (g)-(h) Detection results of the jetski and surrounding swimmers in the SeaDronesSeeV2 dataset under different UAV viewpoints. Despite the change in perspective from a left-side oblique angle (g) to a rightside oblique angle (h), the jetski is consistently detected with high confidence and well-aligned bounding boxes.

5) Evaluation of DCA Stability Under UAV Viewpoint Changes: To evaluate the robustness of the DCA module to changes in UAV shooting angles, we visualize detection results under varying viewpoints using examples from the VisDrone-DET and SeaDronesSeeV2 datasets (see Fig. 12). The results show consistent localization performance, with bounding boxes remaining well aligned across moderate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

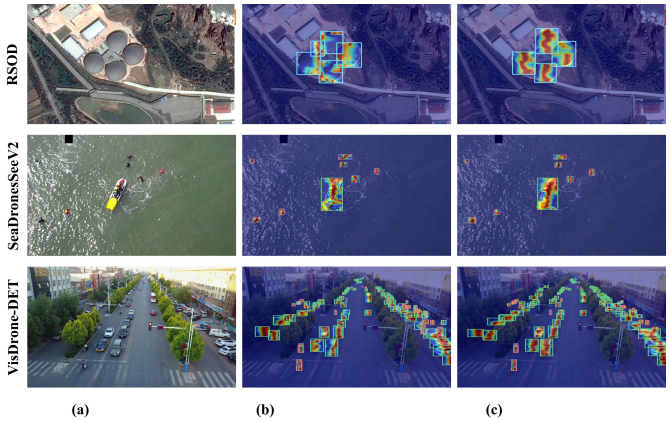


Fig. 13. LayerCAM visualizations of SME-DETR with and without the DCA module on the RSOD, SeaDronesSeeV2, and VisDrone-DET datasets. From left to right: (a) original image, (b) SME-DETR without DCA, and (c) SME-DETR with DCA. These results show that DCA enhances directional texture perception and yields more coherent feature activations across diverse scenes.

viewpoint changes and exhibiting only slight fluctuations in confidence scores.

6) LayerCAM-Based Visualization of DCA Texture Perception: Building on the viewpoint analysis in Section IV-F.5, LayerCAM is employed to visualize how the DCA module modulates internal activations associated with directional texture perception. Fig. 13 compares SME-DETR variants with and without DCA on RSOD, SeaDronesSeeV2, and VisDrone-DET, and shows that DCA reorganizes feature responses, yielding more orientation-aware and boundary-coherent representations.

On RSOD, the variant without DCA exhibits scattered and incoherent activations within the oiltank regions, indicating weak sensitivity to structural textures. With DCA, activations become concentrated and form continuous, direction-aligned patterns across the oiltank surfaces, reflecting improved perception of directional textures. On SeaDronesSeeV2, both variants focus on the main boat region; however, without DCA, the activations are locally concentrated yet remain fragmented along the hull. With DCA, responses become more uniformly distributed and show clearer continuity along the boat's direction, indicating enhanced texture coherence and stability under complex water-surface patterns. On VisDrone-DET, the model without DCA shows activations only over parts of the vans, producing fragmented and direction-inconsistent responses. With DCA, activations become spatially continuous, cover the entire van, and align with its orientation. This behavior is consistent with DCA's design, which strengthens directional texture cues via horizontal and vertical convolutions, thereby achieving coherent structural representations. Overall, the LayerCAM visualizations reveal consistent and coherent activations across diverse scenes, demonstrating that DCA maintains stable texture perception under varying conditions.

7) Failure Case Analysis and Method Limitations: First, the model struggles to detect extremely small, distant, or partially occluded objects. Although the DSA module is embedded at all backbone stages to improve scale adaptiveness, deeper branches suffer from resolution loss, while shallow branches lack semantic abstraction. This dual constraint weakens the retention of fine-grained features under low-contrast or occluded conditions. Second, false detections frequently occur in scenes with repetitive textures or complex structures. The auxiliary BSG module effectively suppresses background-dominated responses and reduces spurious activations. However, these distractors may still be mistakenly recognized as foreground targets in scenarios characterized by highly similar HH patterns, such as wave reflections, urban fences, or industrial cables. Furthermore, the DCA module in the SAA-Backbone improves directional edge modeling but may amplify elongated background responses, increasing confusion with object contours. Finally, the model exhibits class confusion in cluttered environments. Although the EfficientAIFI module enables global context modeling, it lacks explicit category-aware attention. This leads to misclassification in ambiguous scenes, such as swimmers versus boats, particularly under occlusion or pose variation. These limitations are reflected in the failure cases shown in Fig. 14. For

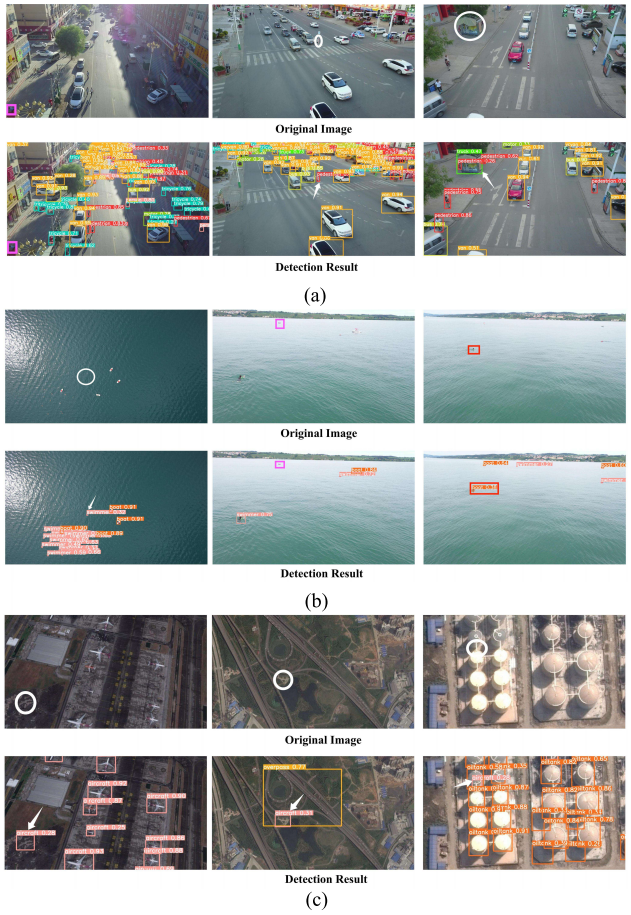


Fig. 14. Presents representative failure cases that expose the inherent limitations of SME-DETR in complex real-world scenarios. (a) Failure cases on the VisDrone-DET dataset. (Left) A pedestrian (purple box) is missed. (Middle & Right) A city fence and a roadside kiosk are falsely detected as a pedestrian and a truck, respectively (white circles and arrows). (b) Failure cases on the SeaDronesSeeV2 dataset. (Left) Water glints are misclassified as swimmers (white circle and arrow). (Middle) A boat (purple box) is missed in detection. (Right) A swimmer is incorrectly classified as a boat (red box). (c) Failure cases on the RSOD dataset. (Left & Middle) Industrial background regions are misclassified as aircraft. (Right) Pipeline connections between oiltanks are incorrectly detected as an aircraft (indicated by white circles and arrows).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

example, the model fails to detect occluded pedestrians in VisDrone-DET, misclassifies industrial structures as airplanes in RSOD, and confuses water glints with swimmers in SeaDronesSeeV2. These limitations point to potential directions for improvement, such as incorporating semantic-aware background suppression, resolution-adaptive detection branches, and fine-grained category discrimination.

V. Conclusion

This article proposes an SME-DETR for small object detection in UAV remote sensing imagery. First, the SAA-Backbone integrates multiscale features adaptively and enhances directional feature representation, effectively overcoming the shortcomings of traditional CNNs in feature extraction. Then, the proposed MDAP integrates spatial, frequency, and channel domain features. This design addresses the limitations of traditional feature pyramids in modeling the global context at high-resolution feature layers. Finally, the constructed EfficientAIFI replaces the complex matrix operations of traditional MHSA with linear element-wise multiplication, thereby maintaining global relational modeling capability while further improving detection accuracy. In addition, an auxiliary BSG is incorporated to suppress background-induced noise, effectively reducing false positives with negligible computational overhead. In quantitative experiments, SME-DETR exhibits exceptional performance on the RSOD, VisDrone-DET, and SeaDronesSeeV2 datasets, particularly outperforming various existing CNN and transformer models in key metrics such as APs, mAP@0.5, and mAP@0.5:0.95. In qualitative experiments, SME-DETR demonstrates superior detection capability by visualizing more precise areas of attention through heatmap visualization, effectively suppressing background interference. Moreover, SME-DETR exhibits faster convergence and more stable accuracy improvements, further validating its advantages in training efficiency and model optimization. Despite the promising results, the failure cases illustrated in Fig. 14 highlight several limitations of SME-DETR. The model struggles with extremely small, occluded, or low-contrast objects due to resolution degradation and insufficient semantic support across network branches. Although the BSG module alleviates part of the background interference, false positives remain challenging in scenes with repetitive textures or structural clutter, and elongated background elements may mislead the directional attention mechanism. In addition, interclass confusion can arise in visually ambiguous environments due to the lack of category-aware attention.

Therefore, future work will focus on: 1) enhancing semantic filtering to suppress background distractors; 2) introducing resolution-adaptive or super-resolution branches to improve tiny object detection; 3) incorporating category-aware attention to reduce interclass confusion; 4) developing orientation-invariant feature modeling (e.g., multidirectional convolution or deformable attention) to improve robustness under extreme UAV shooting angles; and 5) integrating lightweight positional encoding techniques (e.g., relative position bias or rotary embeddings) into EfficientAIFI to enhance its capacity to model spatial relationships, while maintaining its computational efficiency.

References

[1] B. Wang, H. Sui, G. Ma, Y. Zhou, and M. Zhou, "GMODet: A real-time detector for ground-moving objects in optical remote sensing images with regional awareness and semantic–spatial progressive interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5605623.

[2] Y. Xiao, J. Wang, Z. Zhao, B. Jiang, C. Li, and J. Tang, "UAV video vehicle detection: Benchmark and baseline," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5609814.

[3] Z. Liu, D. Zhao, B. Yuan, and Z. Jiang, "RescueADI: Adaptive disaster interpretation in remote sensing images with autonomous agents," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5611814.

[4] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[5] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[6] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[7] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," in *Proc. Comput. Vis. Pattern Recognit.* Berlin, Germany: Springer, 2018, pp. 1–6.

[8] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Oct. 2016, pp. 21–37.

[9] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.

[10] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[11] J. Zhou, Y. Liu, B. Peng, L. Liu, and X. Li, "MaDiNet: Mamba diffusion network for SAR target detection," *IEEE Trans. Circuits Syst. Video Technol.*, early access, May 28, 2025, doi: 10.1109/TCSVT.2025.3574657.

[12] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[13] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[14] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.

[15] S. Ma, Y. Zhang, L. Peng, C. Sun, B. Ding, and Y. Zhu, "OWRT-DETR: A novel real-time transformer network for small-object detection in open-water search and rescue from UAV aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4205313.

[16] X. Zhong et al., "Adaptive deformation-learning and multiscale-integrated network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5611619.

[17] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[18] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[19] C. Li et al., "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.

[20] J. Nie, Y. Pang, S. Zhao, J. Han, and X. Li, "Efficient selective context network for accurate object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 9, pp. 3456–3468, Sep. 2021.

[21] P. Singh, V. K. Verma, P. Rai, and V. P. Namboodiri, "HetConv: Heterogeneous kernel-based convolutions for deep CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4835–4844.

[22] D. Hou, S. Wang, X. Tian, and H. Xing, "An attention-enhanced end-to-end discriminative network with multiscale feature learning for remote sensing image retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8245–8255, 2022.

[23] Z. Han et al., "Detection of standing dead trees after pine wilt disease outbreak with airborne remote sensing imagery by multi-scale spatial attention deep learning and Gaussian kernel approach," *Remote Sens.*, vol. 14, no. 13, p. 3075, Jun. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.

[24] Q. Ran, Q. Wang, B. Zhao, Y. Wu, S. Pu, and Z. Li, "Lightweight oriented object detection using multiscale context and enhanced channel attention in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5786–5795, 2021.

[25] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[26] S. Liu, D. Huang, and Y. Wang, "Learning spatial fusion for single-shot object detection," 2019, *arXiv:1911.09516*.

[27] G. Ghiasi, T.-Y. Lin, and Q. V. Le, "NAS-FPN: Learning scalable feature pyramid architecture for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7036–7045.

[28] H. Zhang, K. Wang, Y. Tian, C. Gou, and F.-Y. Wang, "MFR-CNN: Incorporating multi-scale features and global information for traffic object detection," *IEEE Trans. Veh. Technol.*, vol. 67, no. 9, pp. 8019–8030, Sep. 2018.

[29] S. Woo, S. Hwang, and I. S. Kweon, "StairNet: Top-down semantic aggregation for accurate one shot detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 1093–1102.

[30] C. Deng, M. Wang, L. Liu, Y. Liu, and Y. Jiang, "Extended feature pyramid network for small object detection," *IEEE Trans. Multimedia*, vol. 24, pp. 1968–1979, 2022.

[31] G. Cheng, C. Lang, M. Wu, X. Xie, X. Yao, and J. Han, "Feature enhancement network for object detection in optical remote sensing images," *J. Remote Sens.*, vol. 2021, pp. 1–14, Jan. 2021.

[32] K. Zhang and H. Shen, "Multi-stage feature enhancement pyramid network for detecting objects in optical remote sensing images," *Remote Sens.*, vol. 14, no. 3, p. 579, Jan. 2022.

[33] Z. Zhang, X. Lu, G. Cao, Y. Yang, L. Jiao, and F. Liu, "ViT-YOLO: Transformer-based YOLO for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 2799–2808.

[34] M. Jaderberg et al., "Spatial transformer networks," in *Proc. 29th Annu. Conf. Neural Inf. Process. Syst.*, 2015, pp. 1–9.

[35] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[36] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[37] K. Song, H. Yang, and Z. Yin, "Multi-scale attention deep neural network for fast accurate object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 10, pp. 2972–2985, Oct. 2019.

[38] A. Vaswani, "Attention is all you need," in *Proc. 31st Conf. Neural Inf. Process. Syst.*, 2017, pp. 1–11.

[39] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[40] S. Wang, B. Z. Li, M. Khabsa, H. Fang, and H. Ma, "Linformer: Self-attention with linear complexity," 2020, *arXiv:2006.04768*.

[41] K. Choromanski et al., "Rethinking attention with performers," 2020, *arXiv:2009.14794*.

[42] A. Shaker, M. Maaz, H. Rasheed, S. Khan, M.-H. Yang, and F. S. Khan, "SwiftFormer: Efficient additive attention for transformer-based real-time mobile vision applications," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 17425–17436.

[43] B. Liang, J. Su, K. Feng, Y. Liu, D. Zhang, and W. Hou, "SCDFMixer: Spatial–channel dual-frequency mixer based on satellite optical sensors for remote sensing multiobject detection," *IEEE Sensors J.*, vol. 24, no. 4, pp. 5383–5398, Feb. 2024.

[44] R. Dong, D. Xu, J. Zhao, L. Jiao, and J. An, "Sig-NMS-based faster R-CNN combining transfer learning for small target detection in VHR optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8534–8545, Nov. 2019.

[45] L. Shi, L. Kuang, X. Xu, B. Pan, and Z. Shi, "CANet: Centerness-aware network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603613.

[46] S. Zhou, H. Zhou, and L. Qian, "A multi-scale small object detection algorithm SMA-YOLO for UAV remote sensing images," *Sci. Rep.*, vol. 15, no. 1, pp. 1–15, Mar. 2025.

[47] Z. Hu et al., "Scale-aware pruning framework for remote sensing object detection via multifeature representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614215.

[48] W. Hou et al., "Small object detection method for UAV remote sensing images based on αS-YOLO," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 8984–8994, 2025.

[49] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9650–9660.

[50] B. Du, Y. Huang, J. Chen, and D. Huang, "Adaptive sparse convolutional networks with global context enhancement for faster object detection on drone images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 13435–13444.

[51] F. Yang, H. Fan, P. Chu, E. Blasch, and H. Ling, "Clustered object detection in aerial images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8311–8320.

[52] S. Qi, X. Song, T. Shang, X. Hu, and K. Han, "MSFE-YOLO: An improved YOLOv8 network for object detection on drone view," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[53] N. Battish, D. Kaur, M. Chugh, and S. Poddar, "SDMNet: Spatially dilated multi-scale network for object detection for drone aerial imagery," *Image Vis. Comput.*, vol. 150, pp. 1–11, Oct. 2024.

[54] R. Zhang, Z. Shao, X. Huang, J. Wang, Y. Wang, and D. Li, "Adaptive dense pyramid network for object detection in UAV imagery," *Neurocomputing*, vol. 489, pp. 377–389, Jun. 2022.

[55] C. Yang, Z. Huang, and N. Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13668–13677.

[56] H. Zhang, K. Liu, Z. Gan, and G.-N. Zhu, "UAV-DETR: Efficient end-to-end object detection for unmanned aerial vehicle imagery," 2025, *arXiv:2501.01855*.

[57] Y. Liu, Y. Wu, and Y. Yuan, "DMR-RTDETR: A multiscale and context-aware measurement framework for quantitative characterization of photovoltaic module thermal anomalies with uncertainty analysis," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–14, 2025.

[58] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 107984–108011.

[59] Y. Chen et al., "Accurate leukocyte detection based on deformable-DETR and multi-level feature fusion for aiding diagnosis of blood diseases," *Comput. Biol. Med.*, vol. 170, pp. 1–15, Mar. 2024.

[60] G. Jocher et al., 2023, "Ultralytics/YOLOv5: V7.0-YOLOv5 SOTA realtime instance segmentation (2022)," *Zenodo*.

[61] X. Zhu, S. Lyu, X. Wang, and Q. Zhao, "TPH-YOLOv5: Improved YOLOv5 based on transformer prediction head for object detection on drone-captured scenarios," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 2778–2788.

[62] T. Qiu, W. Zeng, and C. Chen, "FFDFA-YOLO: An enhanced YOLOv8 model for UAV-assisted maritime rescue missions," in *Proc. 9th Int. Symp. Comput. Inf. Process. Technol. (ISCIPT)*, May 2024, pp. 60–63.

[63] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.

[64] W. Wu et al., "EUAVDet: An efficient and lightweight object detector for UAV aerial images with an edge-based computing platform," *Drones*, vol. 8, no. 6, pp. 1–23, Jun. 2024.

[65] B. Zhao and R. Song, "Enhancing two-stage object detection models via data-driven anchor box optimization in UAV-based maritime SAR," *Sci. Rep.*, vol. 14, no. 1, p. 4765, Feb. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:49:23 UTC from IEEE Xplore. Restrictions apply.